German University in Cairo (GUC)
Faculty of Media Engineering and Technology

LearnLoop

AI-Powered Personalized Assessment & Learning Gap Detection for Children

Milestone 2: Persona & Task Analysis

CSEN 909 Human-Computer Interaction · Winter 2026 · Dr. Aya Salama

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Introduction

LearnLoop is a web-based, AI-assisted assessment system for children in Grades 1–5. Students upload scanned exams, OCR extracts their answers, AI grades them and detects learning gaps and red flags, and the teacher verifies and confirms every result before students or parents see it.

Our Milestone 1 review found that misread handwriting causes 23% of grading errors in automated handwritten-exam grading, yet existing systems have no step where a teacher checks the extracted text. LearnLoop's unique feature, Confidence-Guided Verification (CGV), directs the teacher's attention by OCR confidence, learns each student's handwriting, reports OCR problems class-wide, and shows how much was checked by hand versus auto-accepted.

This report presents the Milestone 2 deliverables in the order requested:

1. Diagrammatic Hierarchical Task Analysis (HTA)
2. User personas
3. All features (functional requirements)
4. Low-fidelity wireframes of the key screens
5. Interview with a teacher

Followed by our AI-use statement and AI chat history. All diagrams are also available as view-only Figma links, given next to each section, in case any detail is hard to read in the PDF.

Links (view only)

- Hierarchical Task Analysis (FigJam): LearnLoop – HTA
- User personas (Figma): LearnLoop – Personas
- Low-fidelity wireframes (Figma): LearnLoop – Wireframes

1. Hierarchical Task Analysis (HTA)

We analysed six primary tasks across the student, teacher and parent views. Each HTA follows the same conventions:

- The overall goal is numbered 0, subtasks 1, 2, 3… and sub-steps 1.1, 1.2… (two to three levels).
- A plan is written next to every parent box, giving the order of steps and the conditions ("if… then…") for decisions and loops.
- Steps are interface-independent: they describe what the user wants to achieve, not buttons or screens. Interface detail is left for the Milestone 3 hierarchical task design (HTD).
- Purple boxes mark steps that belong to Confidence-Guided Verification (CGV), our unique feature.

Each HTA is shown in full, followed by enlarged sections so the text stays readable. The original board is available here: LearnLoop – HTA (FigJam)

HTA 1 – Student uploads a scanned exam

Student view

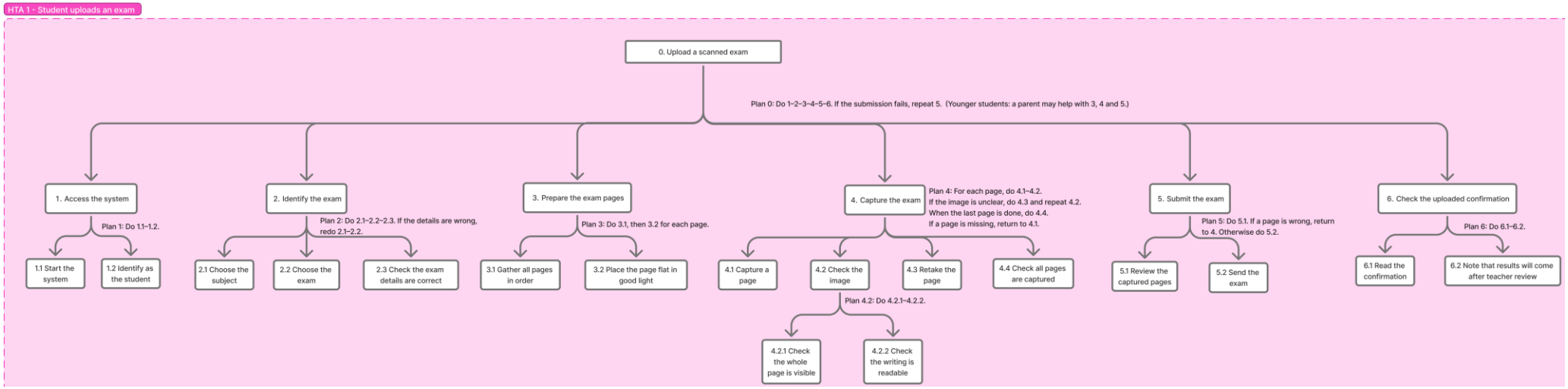
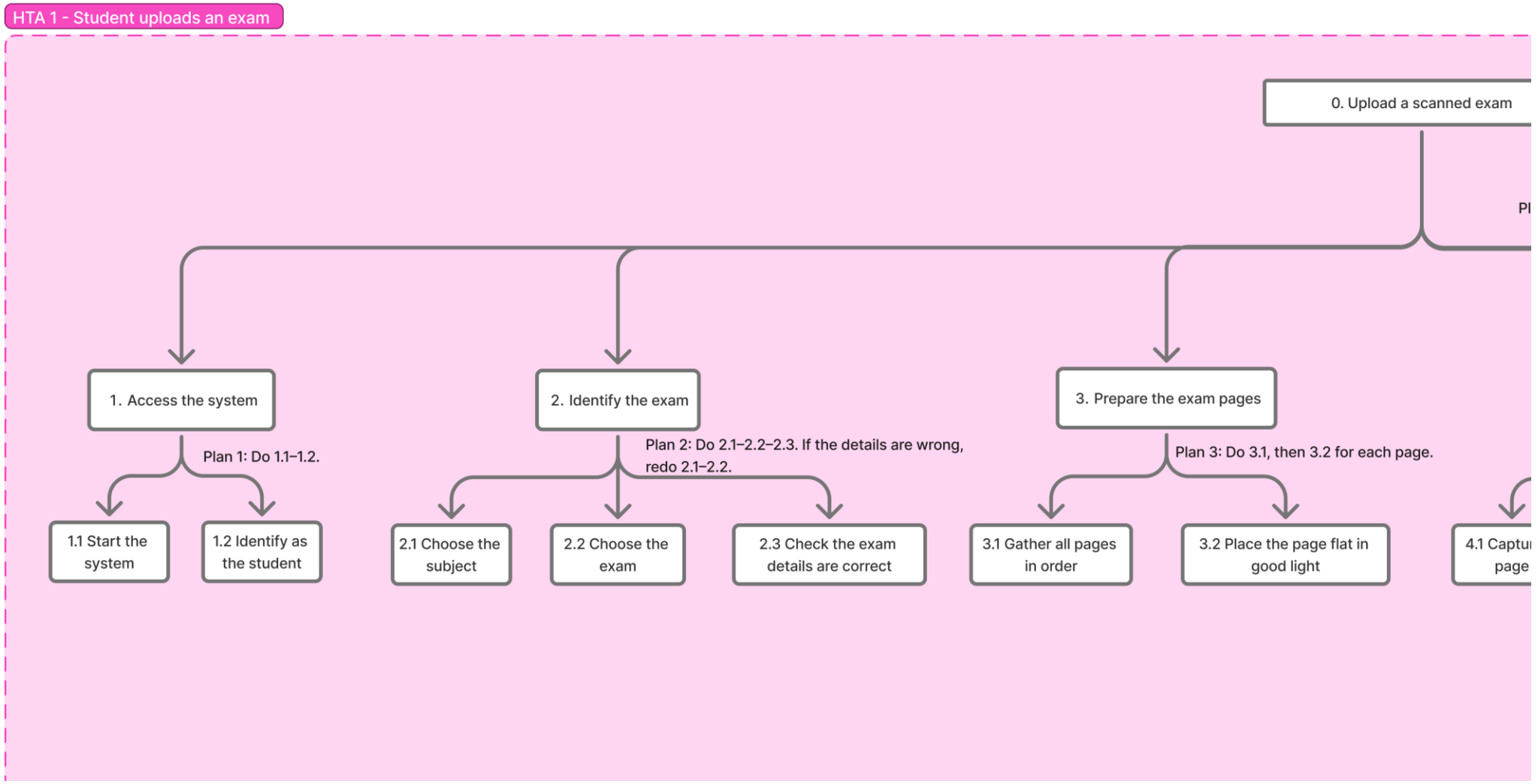


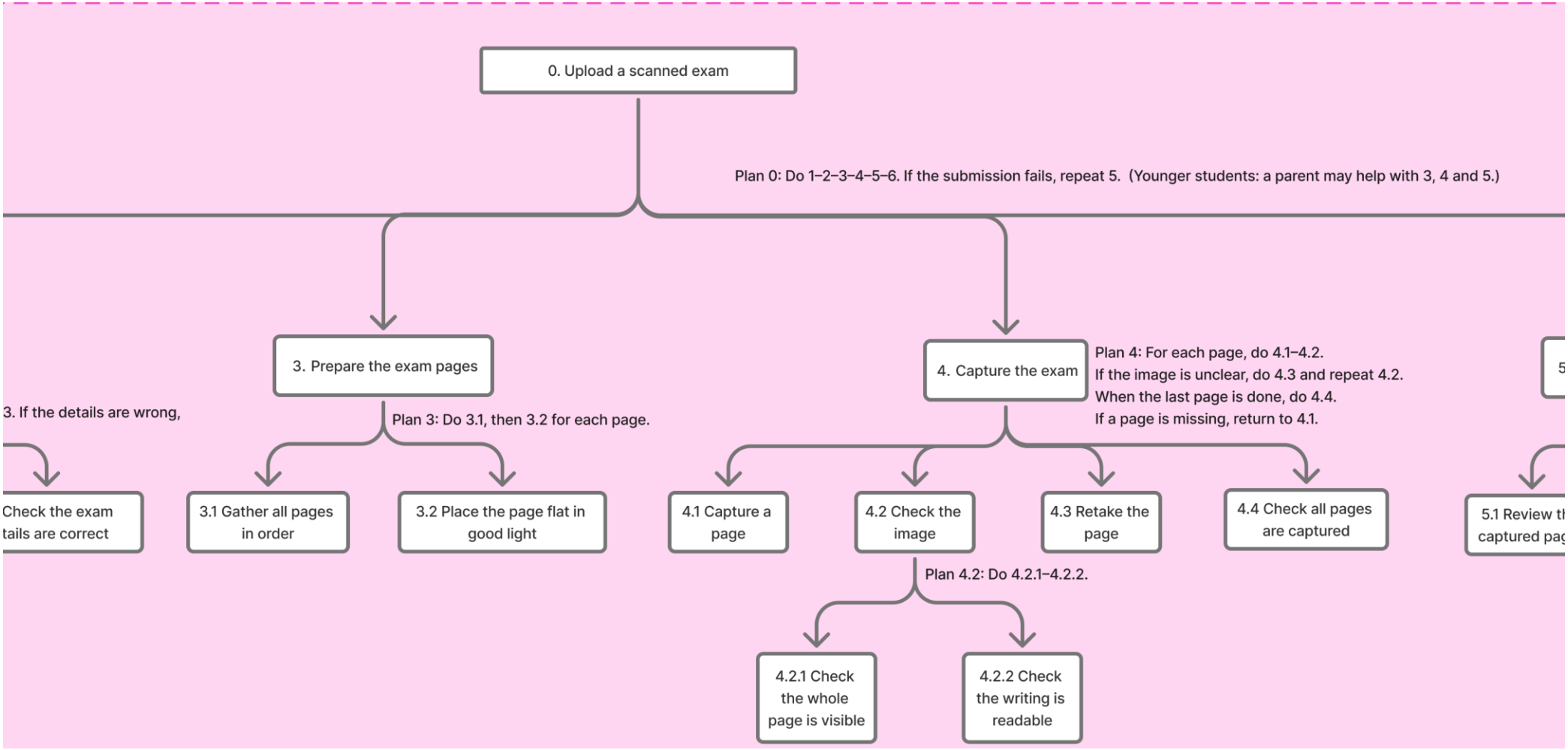
Figure 1.0 – HTA 1 in full

HTA 1 – Student uploads a scanned exam (enlarged, part 1 of 3)



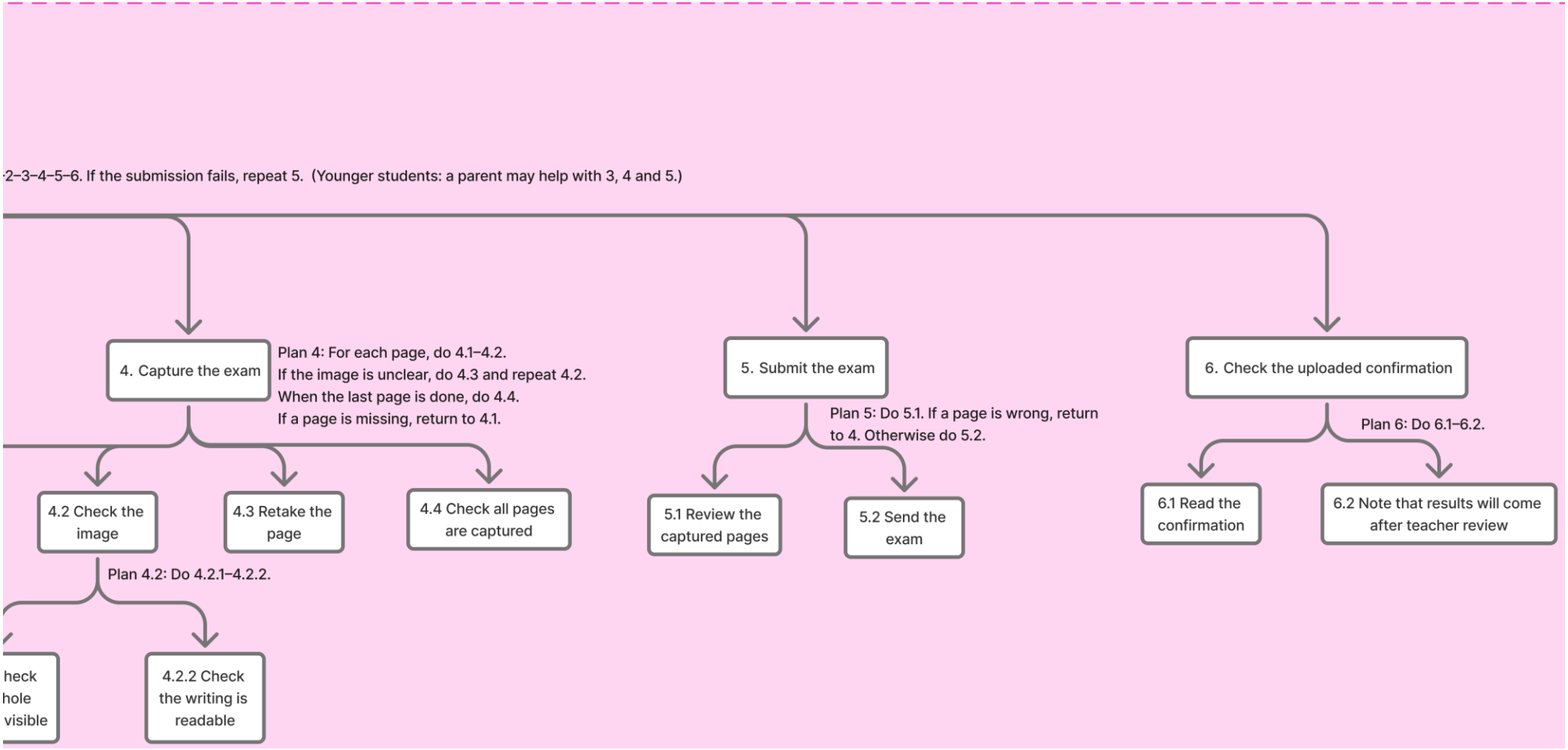
HTA 1, part 1 of 3 (read left to right; parts overlap slightly)

HTA 1 – Student uploads a scanned exam (enlarged, part 2 of 3)



HTA 1, part 2 of 3 (read left to right; parts overlap slightly)

HTA 1 – Student uploads a scanned exam (enlarged, part 3 of 3)



HTA 1, part 3 of 3 (read left to right; parts overlap slightly)

HTA 2 – Teacher verifies OCR text (CGV)

Teacher view · purple boxes are CGV steps

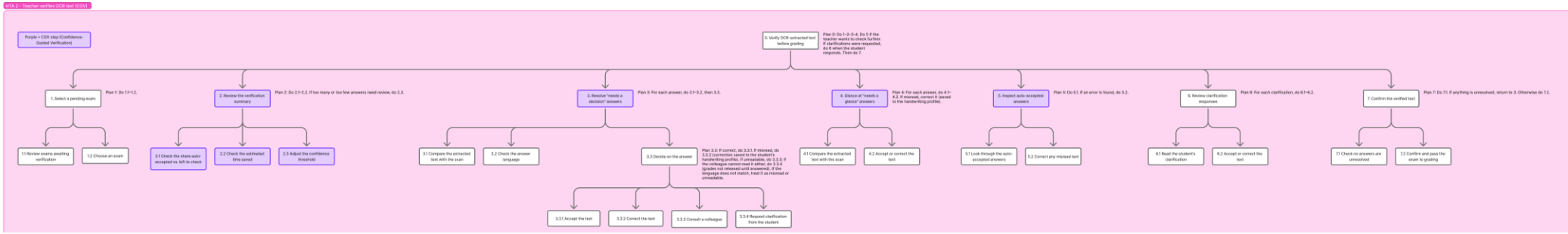
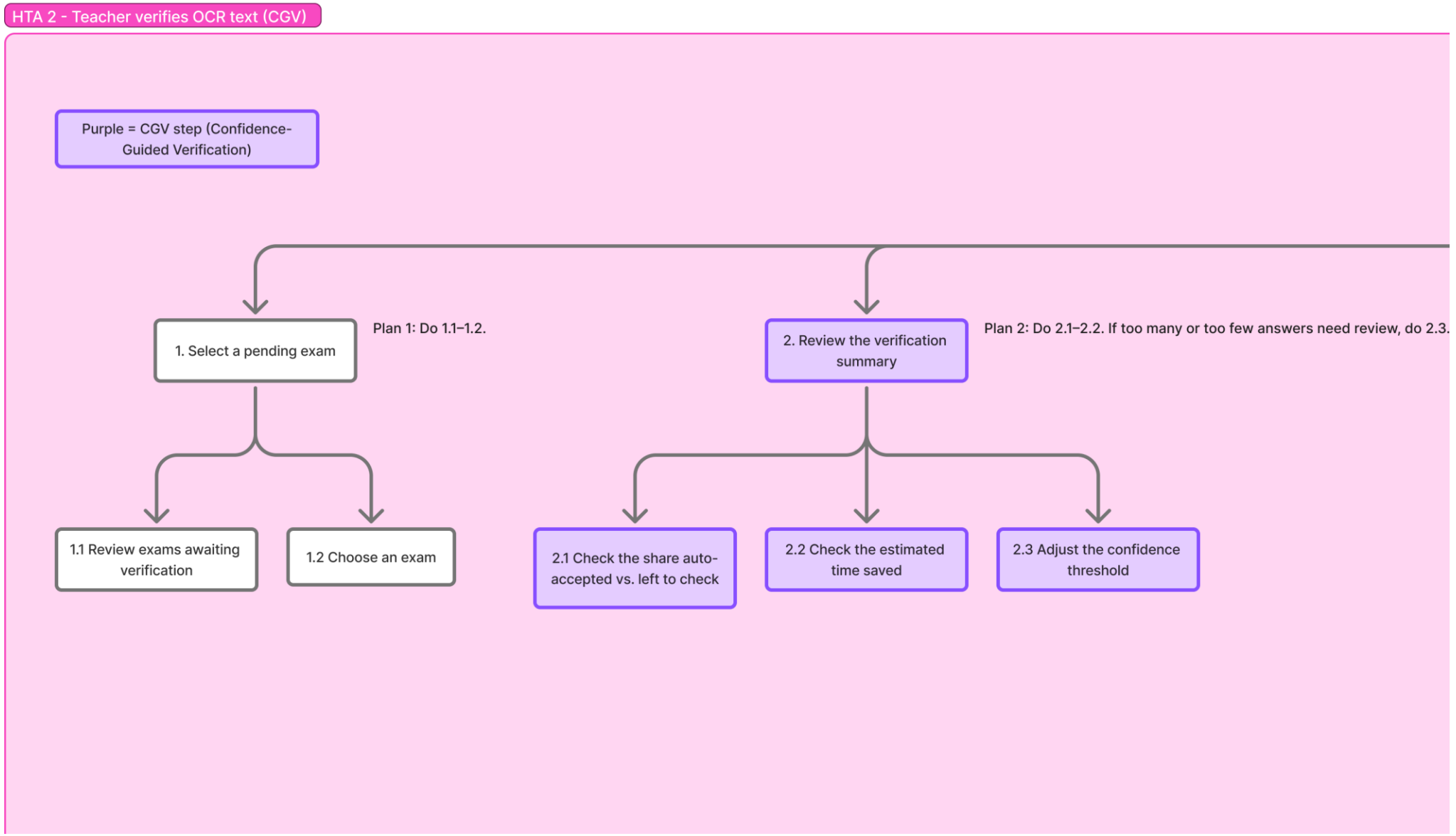


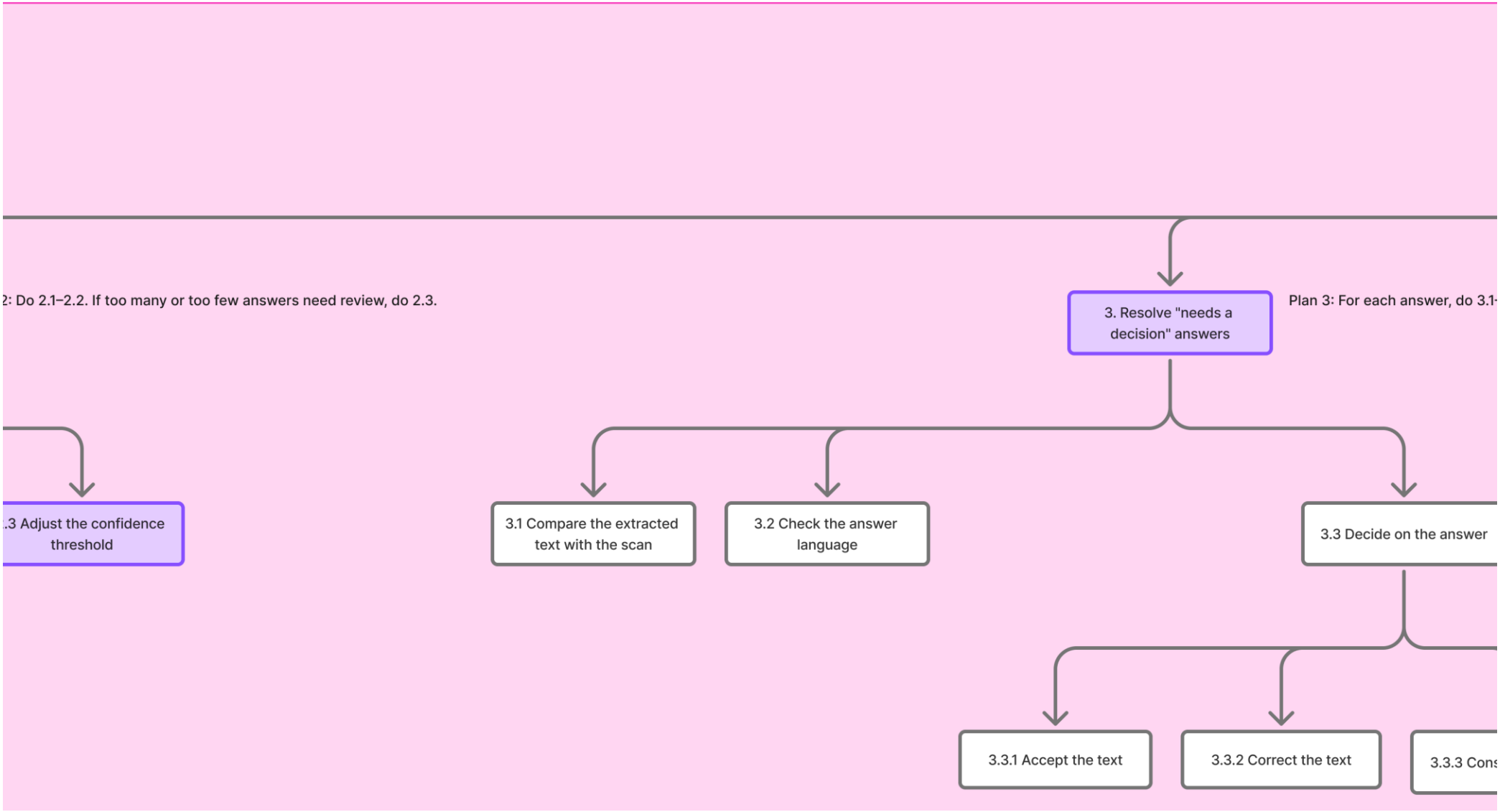
Figure 2.0 – HTA 2 in full

HTA 2 – Teacher verifies OCR text (CGV) (enlarged, part 1 of 5)



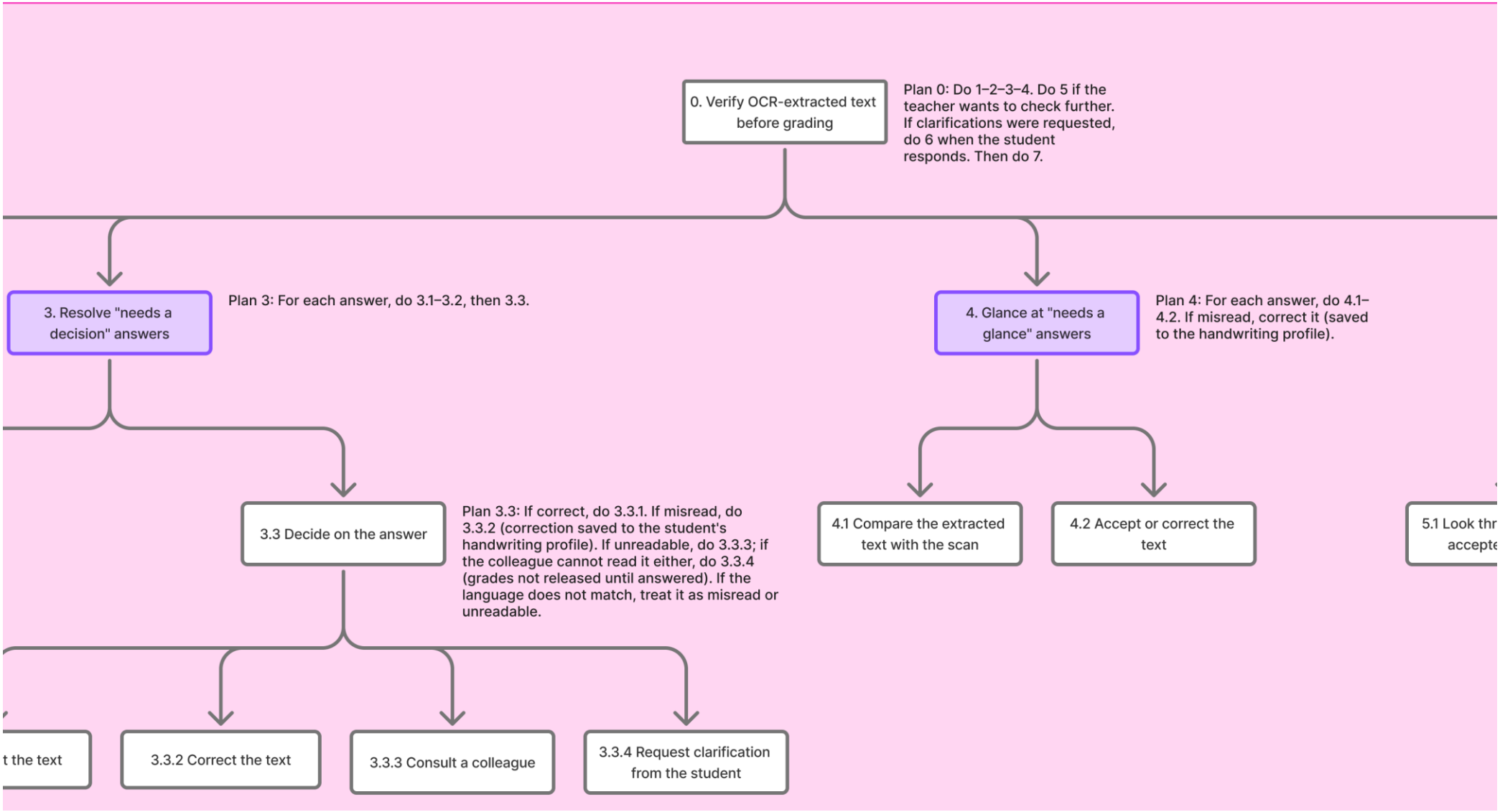
HTA 2, part 1 of 5 (read left to right; parts overlap slightly)

HTA 2 – Teacher verifies OCR text (CGV) (enlarged, part 2 of 5)



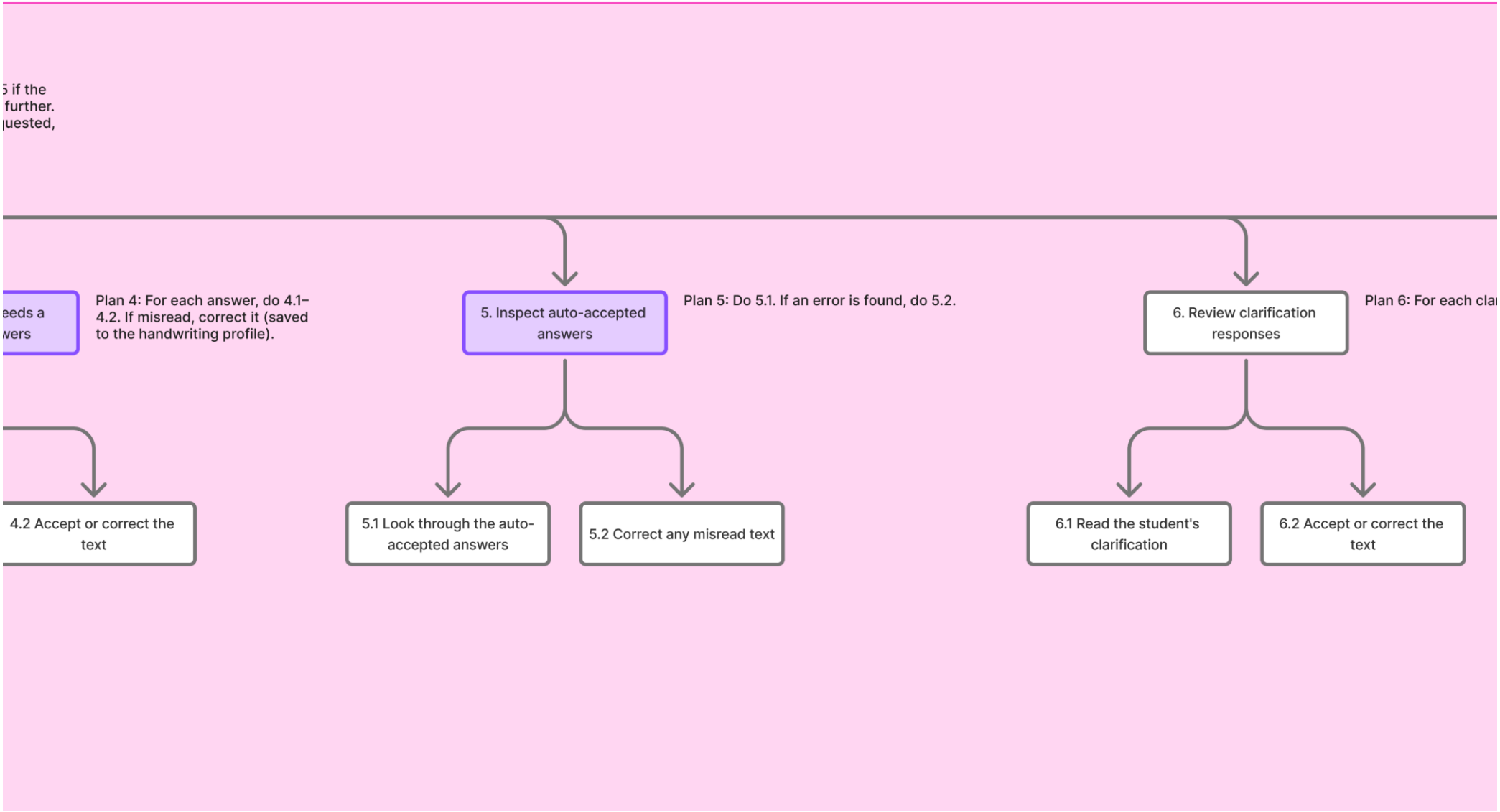
HTA 2, part 2 of 5 (read left to right; parts overlap slightly)

HTA 2 – Teacher verifies OCR text (CGV) (enlarged, part 3 of 5)



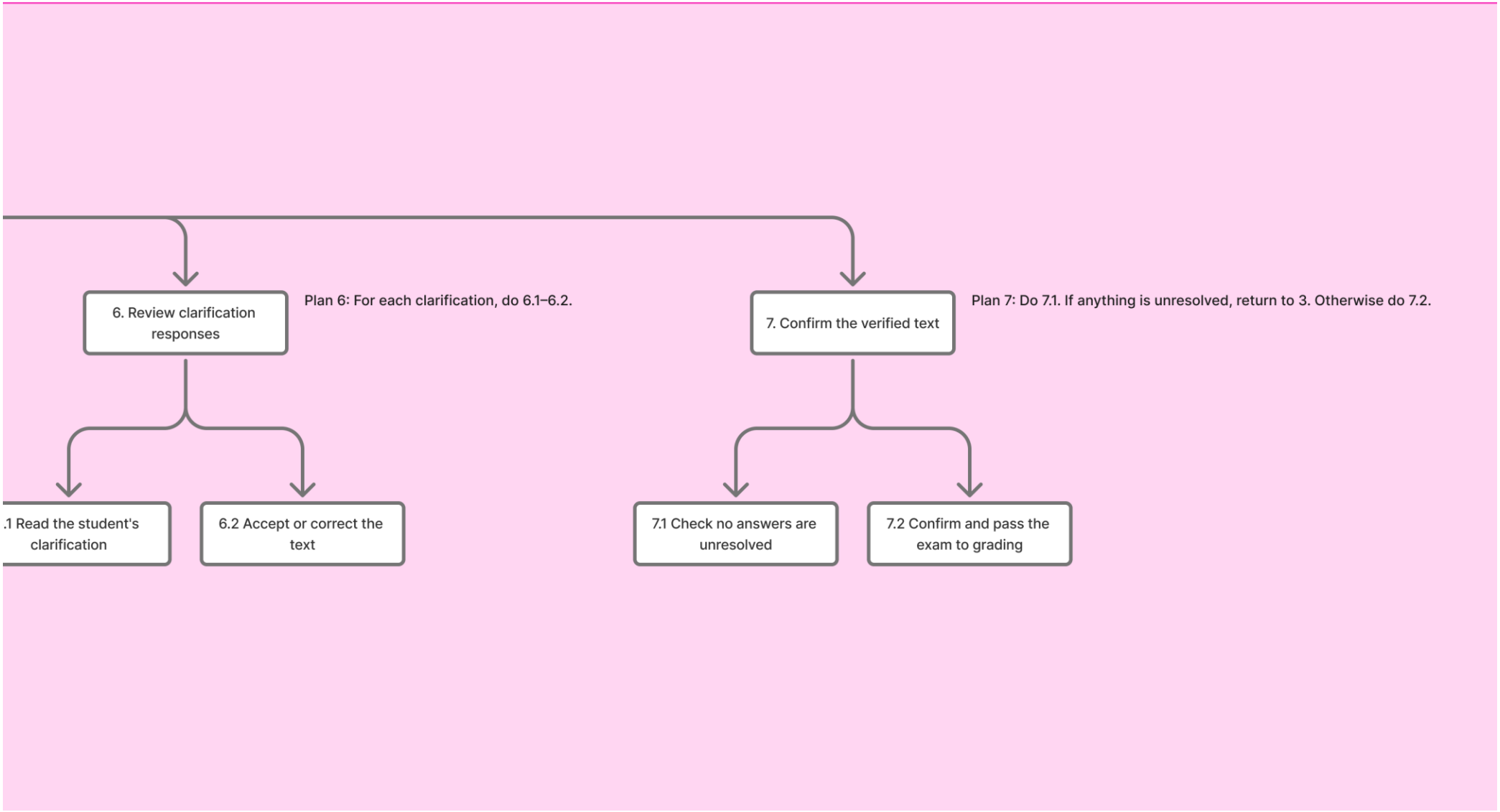
HTA 2, part 3 of 5 (read left to right; parts overlap slightly)

HTA 2 – Teacher verifies OCR text (CGV) (enlarged, part 4 of 5)



HTA 2, part 4 of 5 (read left to right; parts overlap slightly)

HTA 2 – Teacher verifies OCR text (CGV) (enlarged, part 5 of 5)



HTA 2, part 5 of 5 (read left to right; parts overlap slightly)

HTA 3 – Teacher confirms grade, learning gaps and red flags

Teacher view · purple box 2.2 is CGV

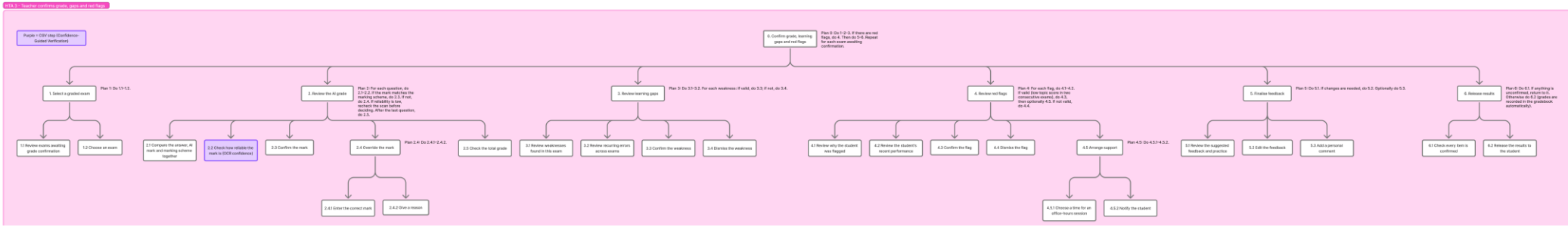
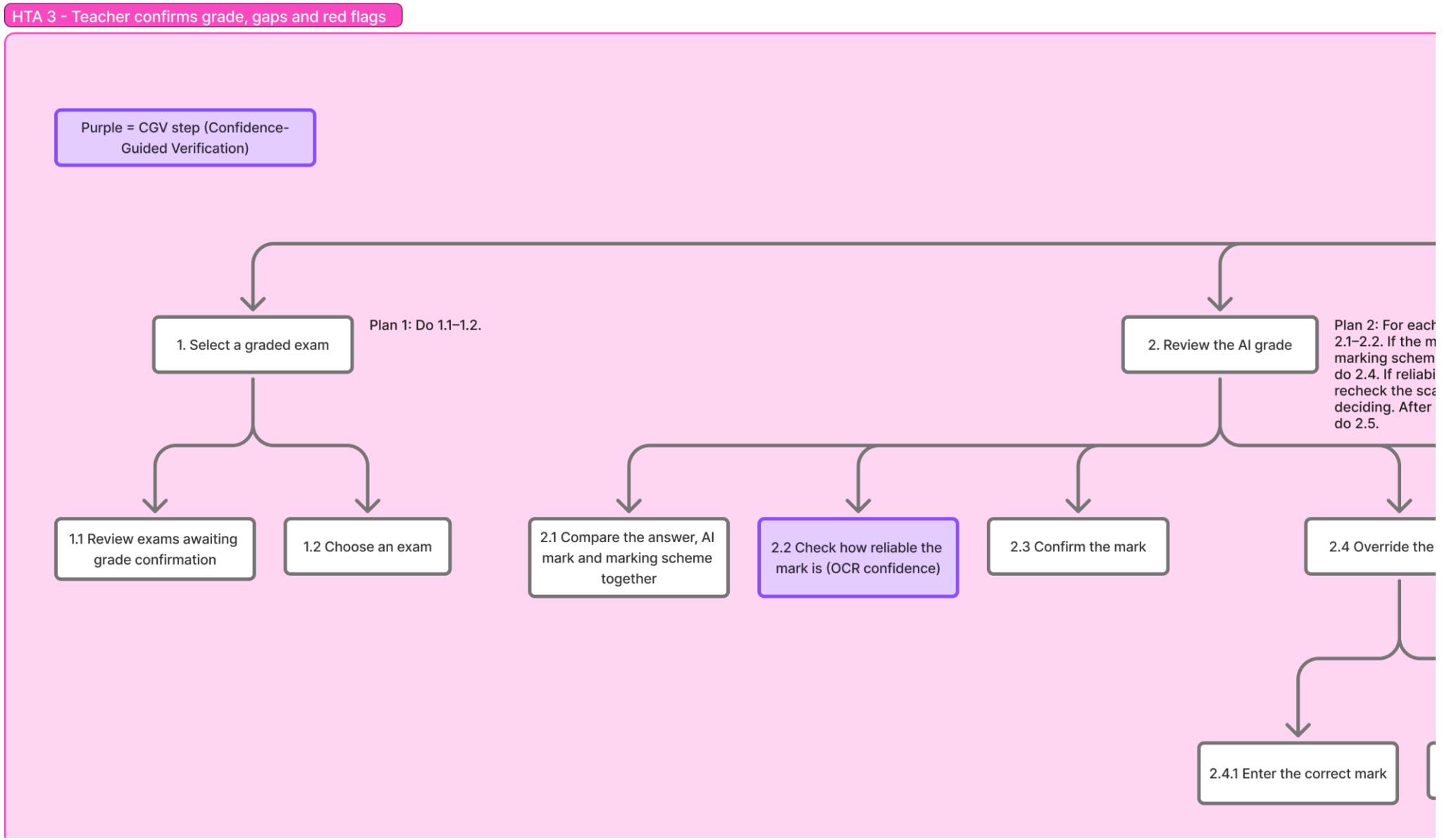


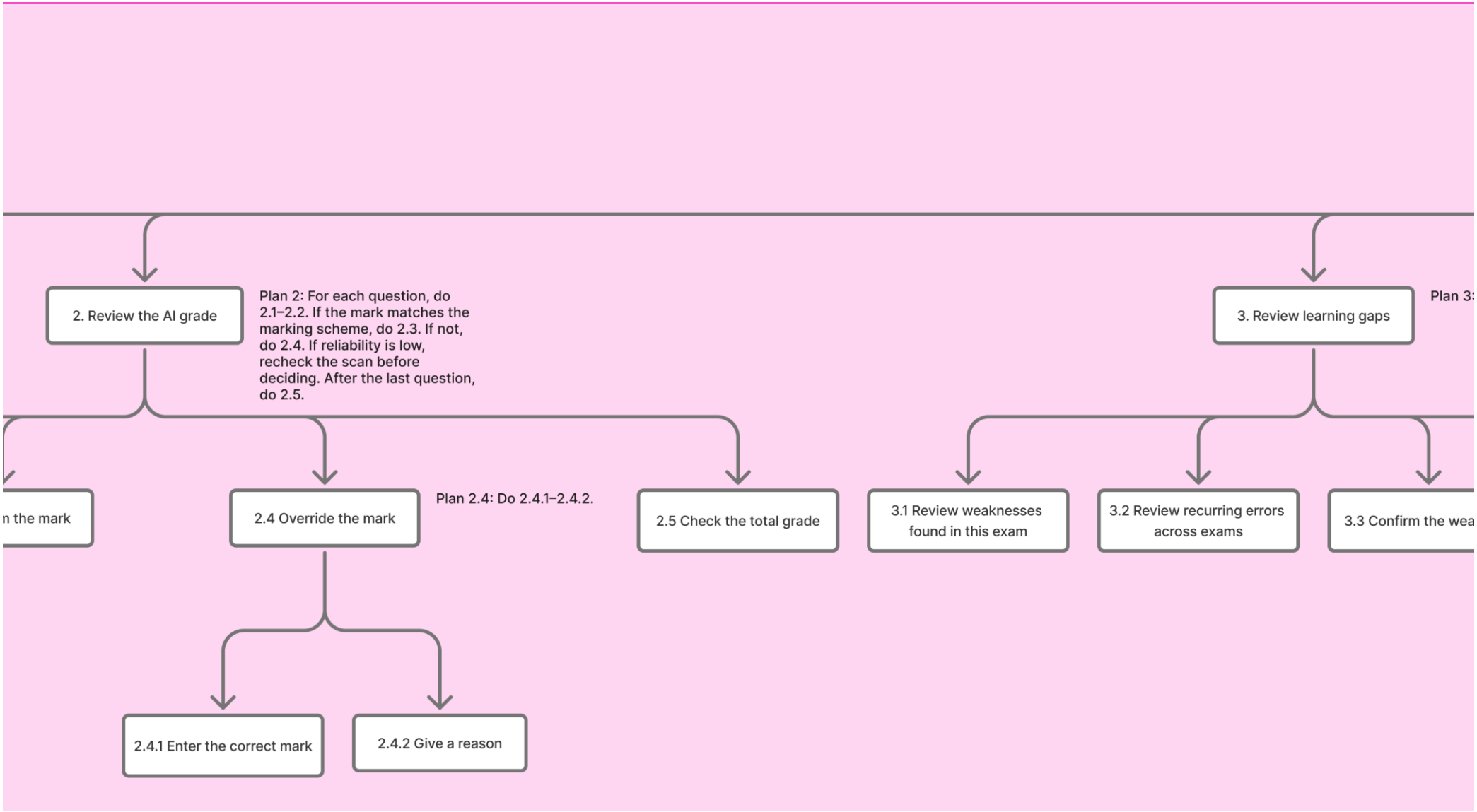
Figure 3.0 – HTA 3 in full

HTA 3 – Teacher confirms grade, learning gaps and red flags (enlarged, part 1 of 5)



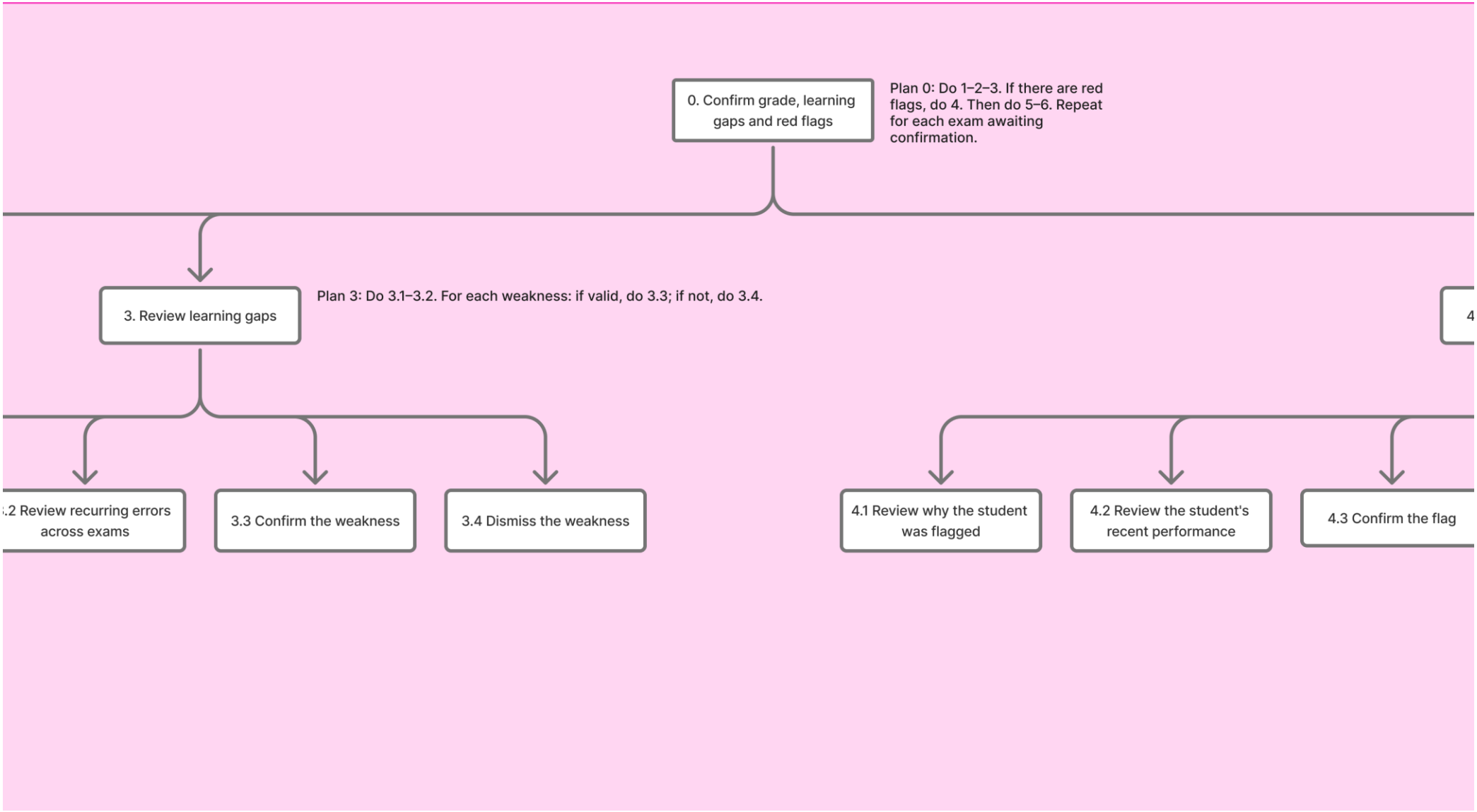
HTA 3, part 1 of 5 (read left to right; parts overlap slightly)

HTA 3 – Teacher confirms grade, learning gaps and red flags (enlarged, part 2 of 5)



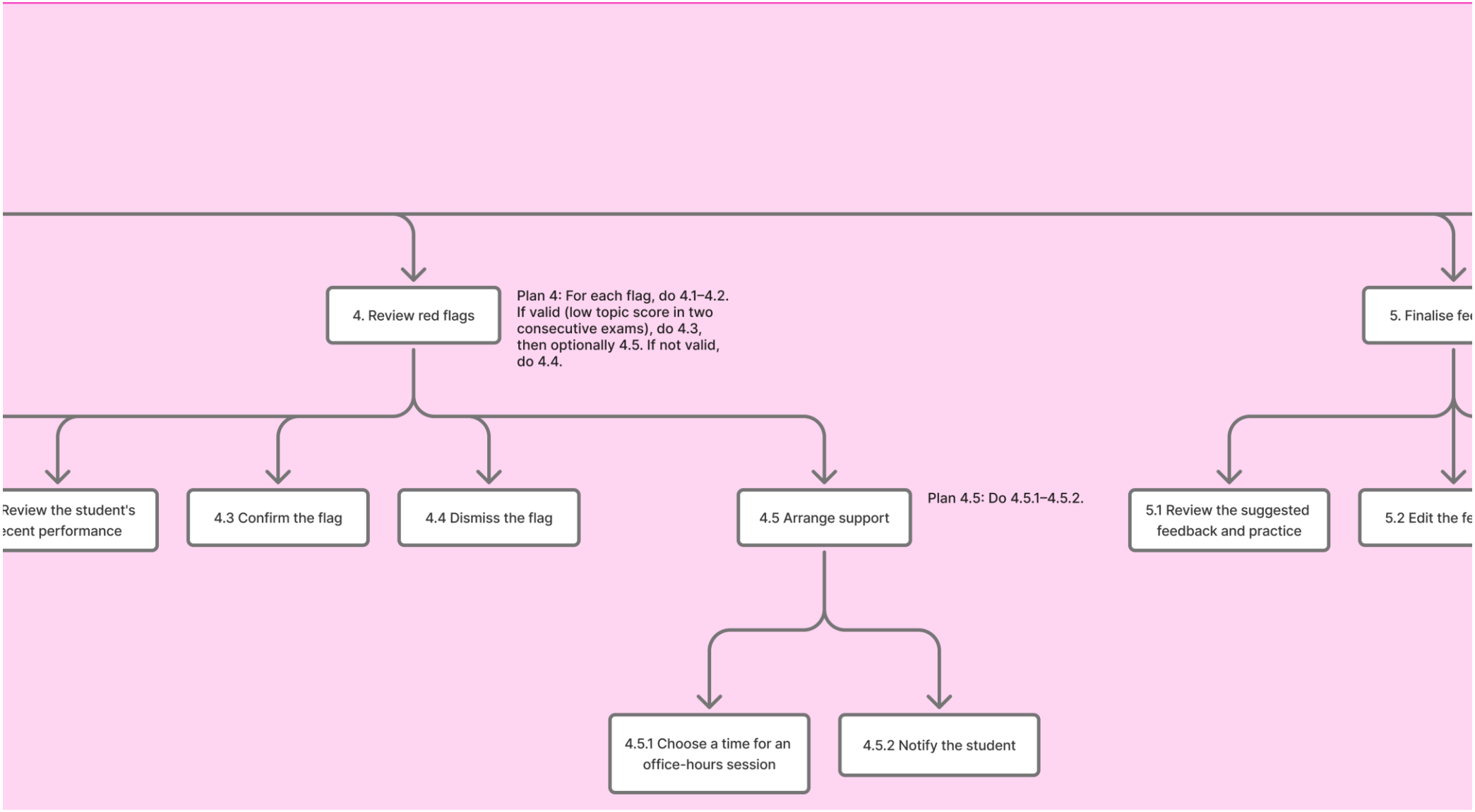
HTA 3, part 2 of 5 (read left to right; parts overlap slightly)

HTA 3 – Teacher confirms grade, learning gaps and red flags (enlarged, part 3 of 5)



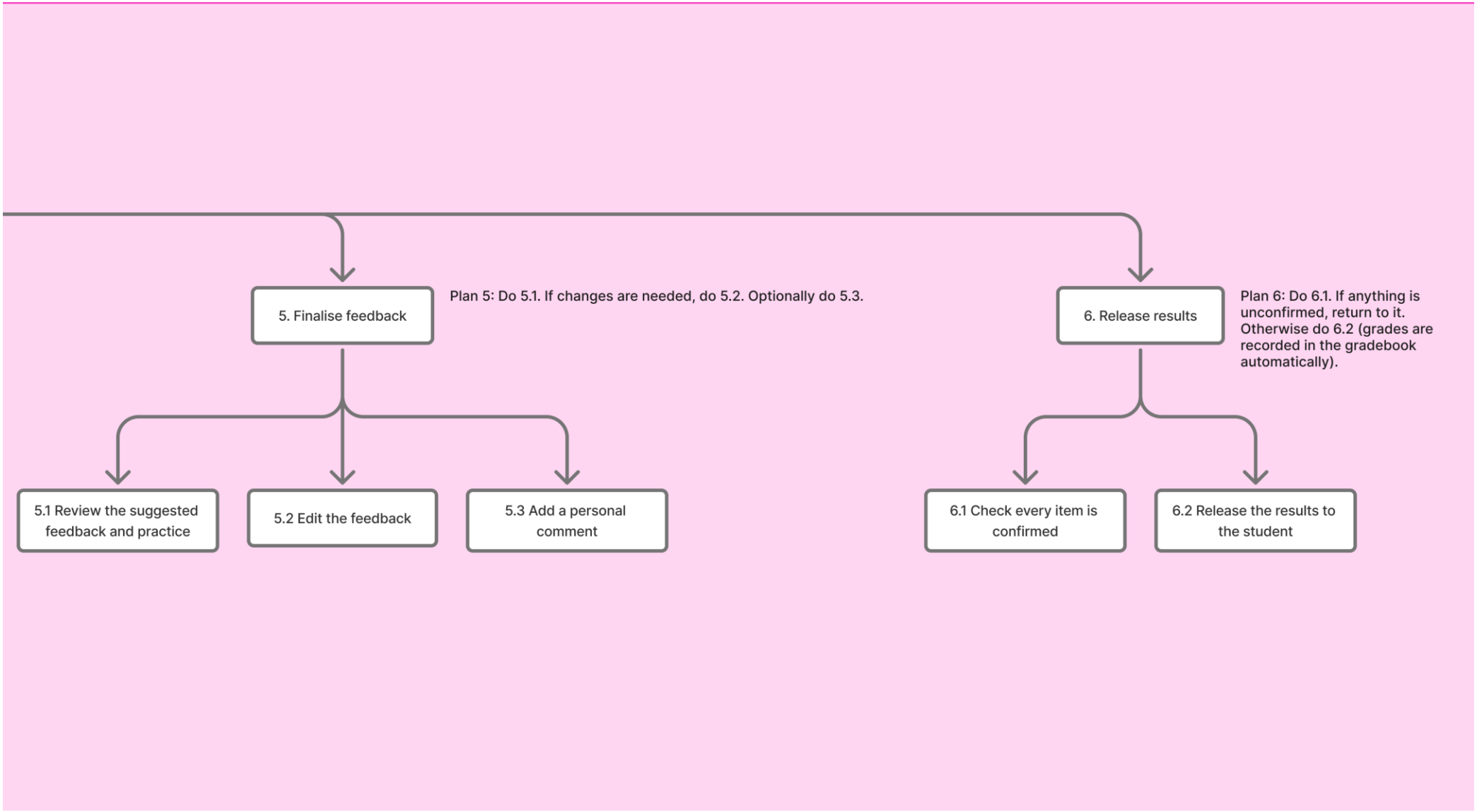
HTA 3, part 3 of 5 (read left to right; parts overlap slightly)

HTA 3 – Teacher confirms grade, learning gaps and red flags (enlarged, part 4 of 5)



HTA 3, part 4 of 5 (read left to right; parts overlap slightly)

HTA 3 – Teacher confirms grade, learning gaps and red flags (enlarged, part 5 of 5)



HTA 3, part 5 of 5 (read left to right; parts overlap slightly)

HTA 4 – Student views results and tracks progress

Student view

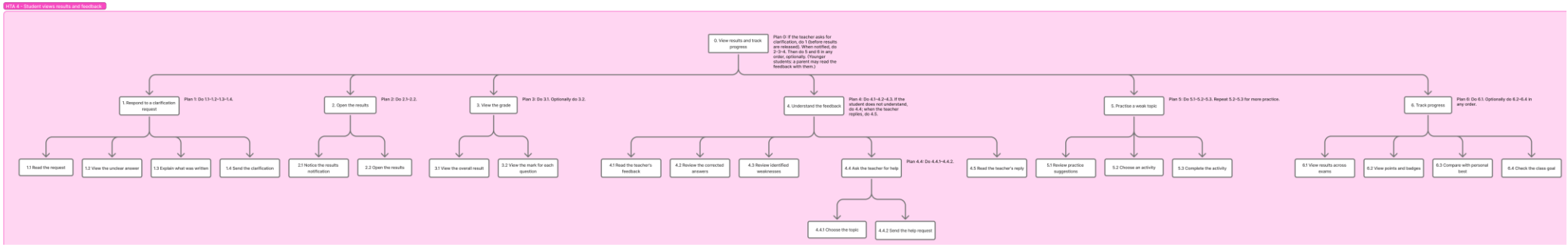
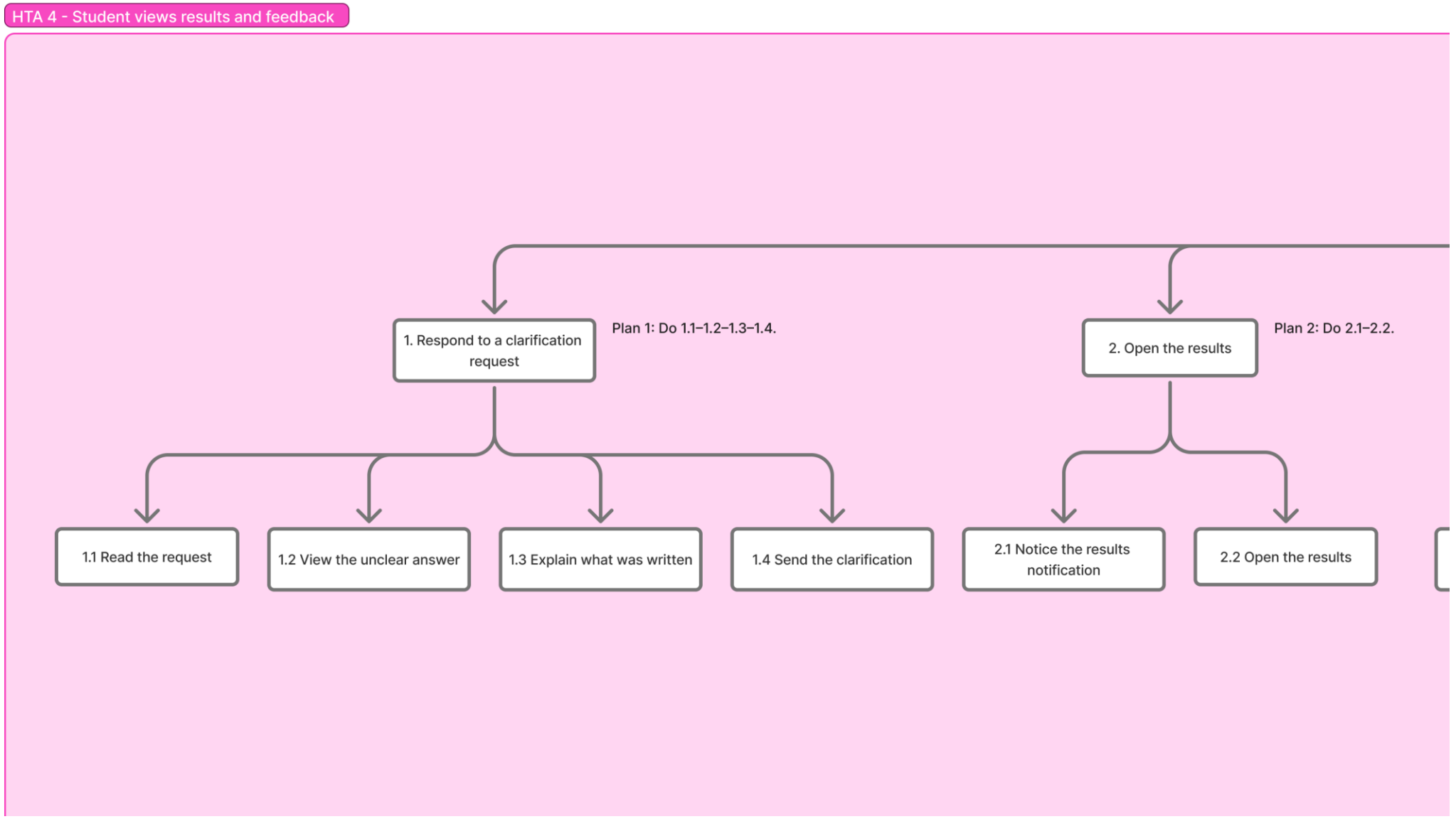


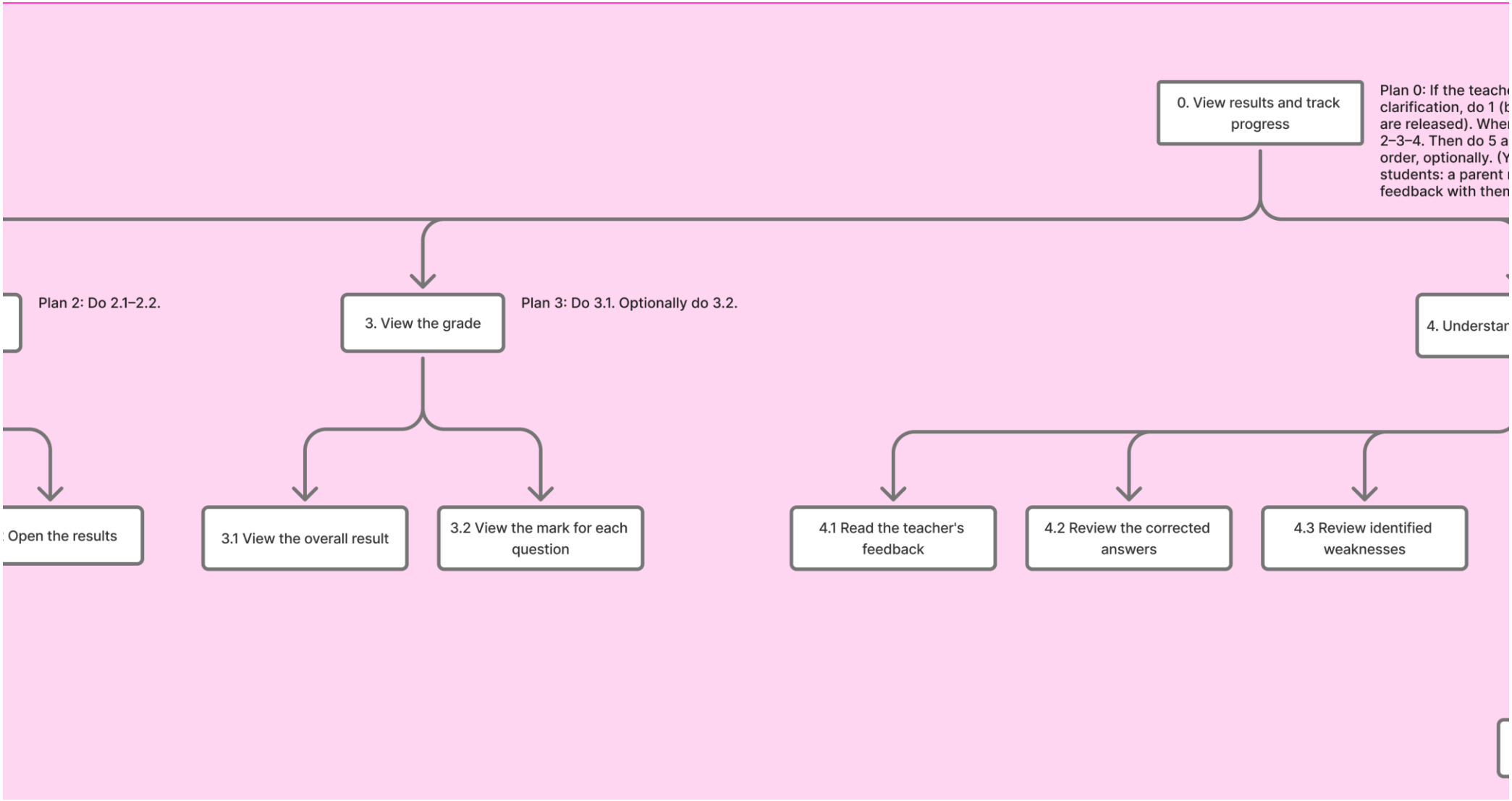
Figure 4.0 – HTA 4 in full

HTA 4 – Student views results and tracks progress (enlarged, part 1 of 4)



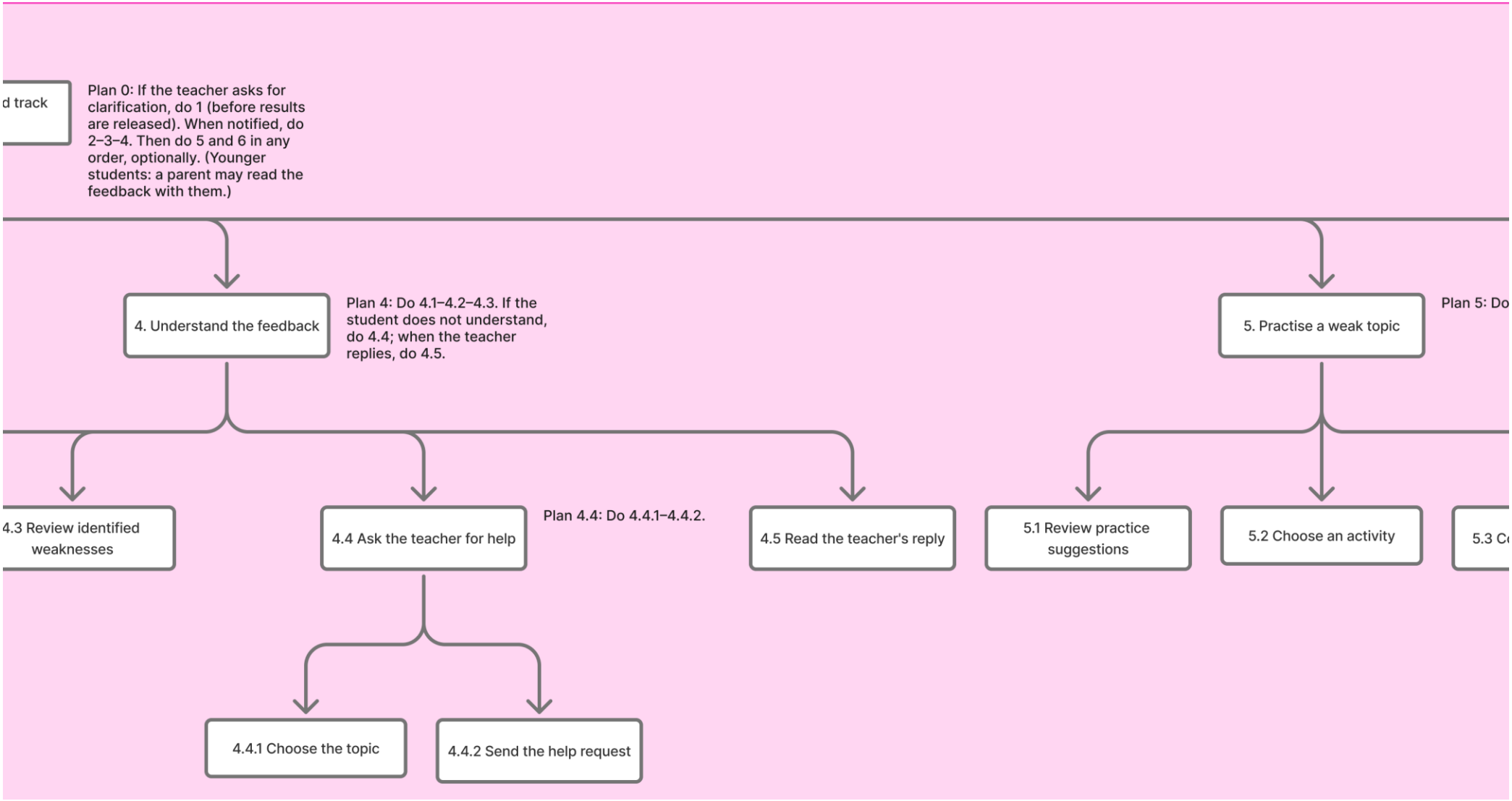
HTA 4, part 1 of 4 (read left to right; parts overlap slightly)

HTA 4 – Student views results and tracks progress (enlarged, part 2 of 4)



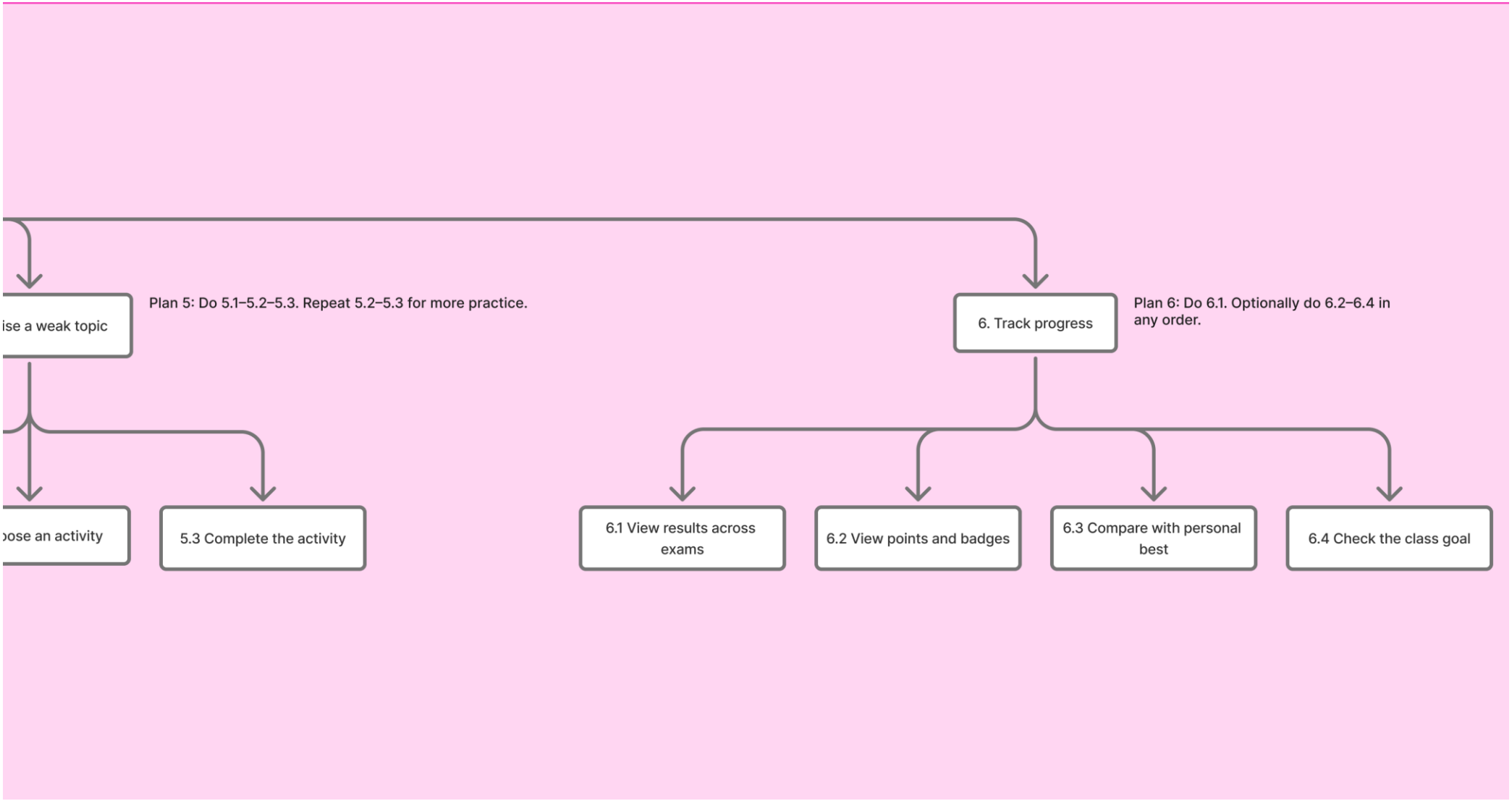
HTA 4, part 2 of 4 (read left to right; parts overlap slightly)

HTA 4 – Student views results and tracks progress (enlarged, part 3 of 4)



HTA 4, part 3 of 4 (read left to right; parts overlap slightly)

HTA 4 – Student views results and tracks progress (enlarged, part 4 of 4)



HTA 4, part 4 of 4 (read left to right; parts overlap slightly)

HTA 5 – Teacher acts on a red flag

Teacher view

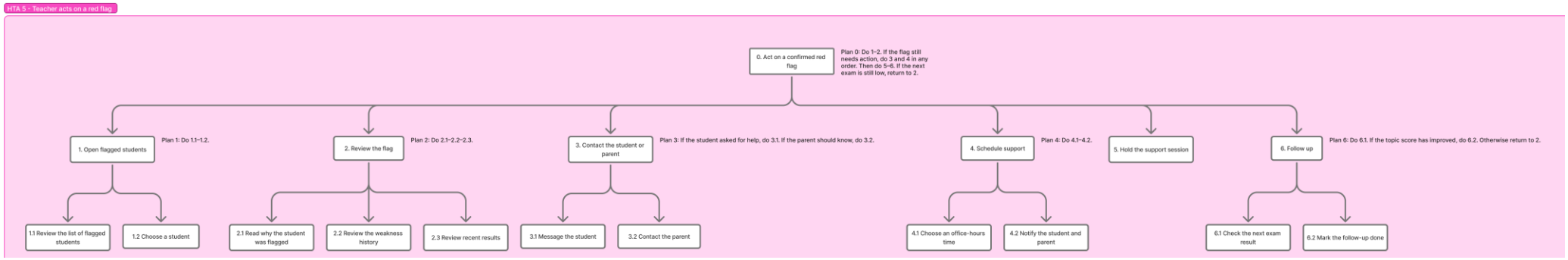
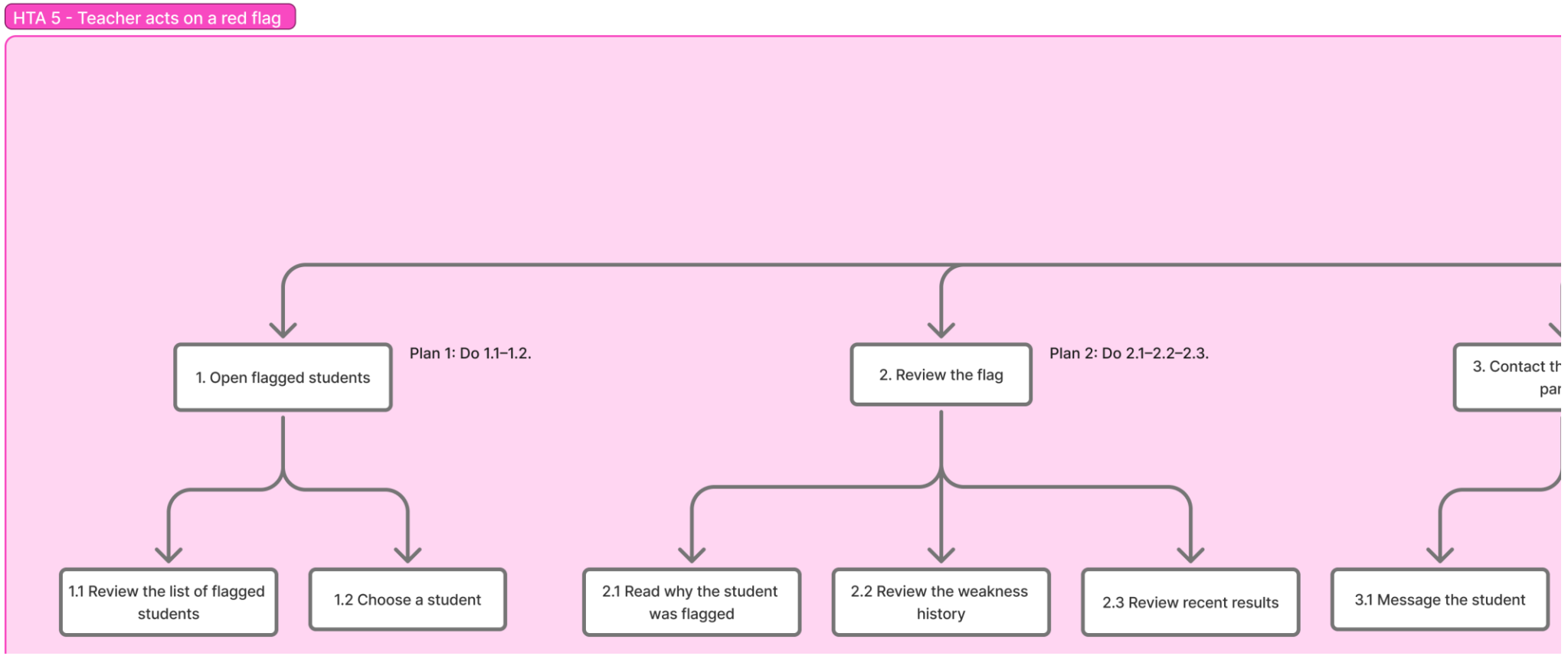


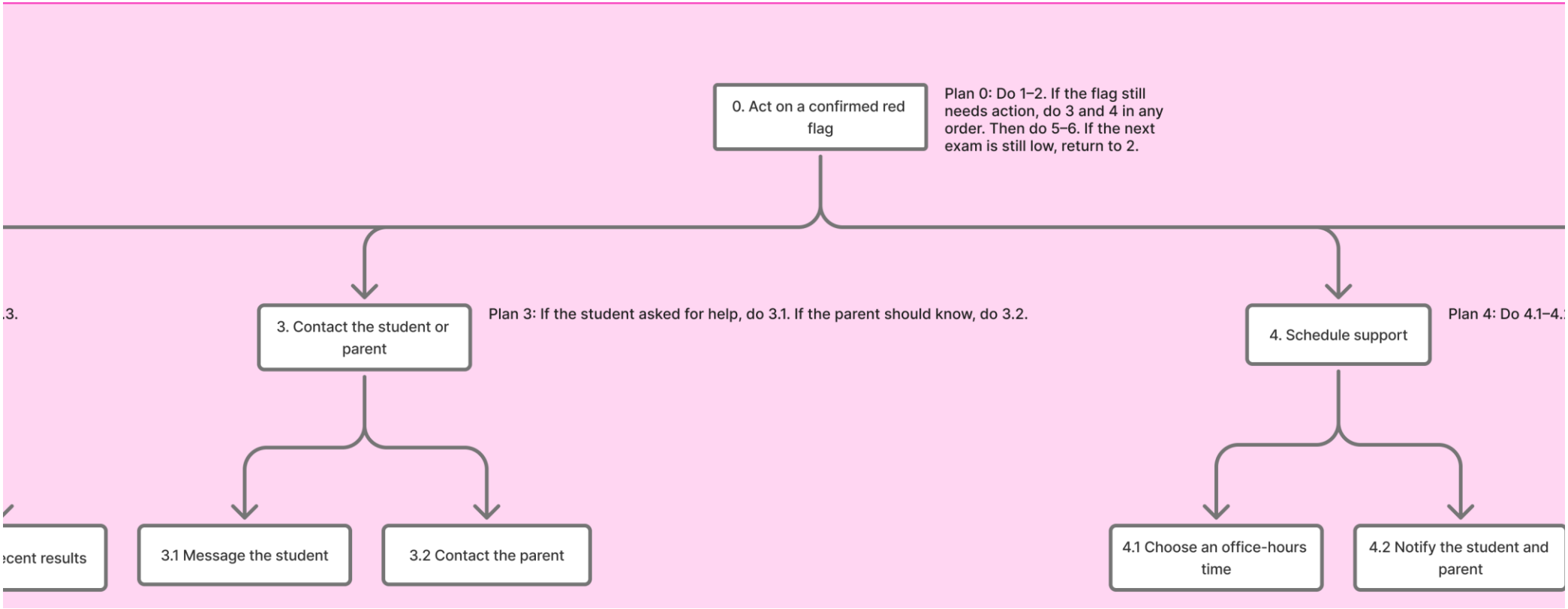
Figure 5.0 – HTA 5 in full

HTA 5 – Teacher acts on a red flag (enlarged, part 1 of 3)



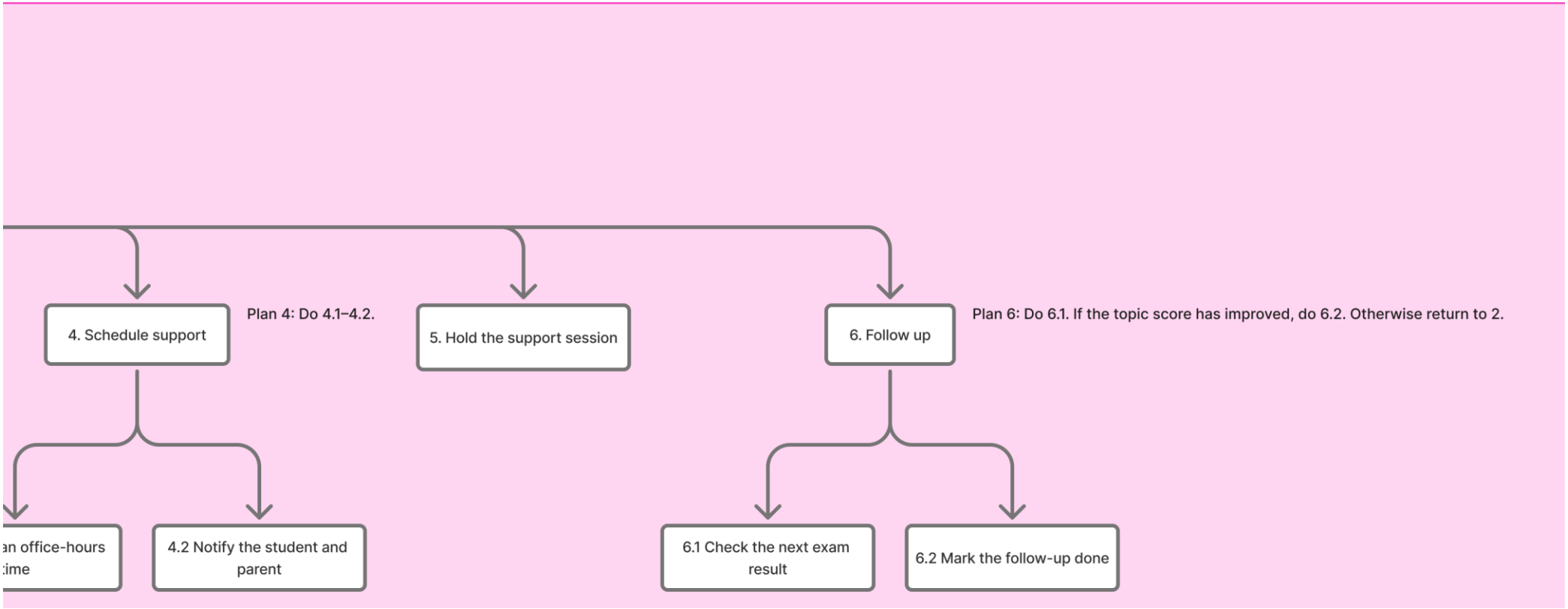
HTA 5, part 1 of 3 (read left to right; parts overlap slightly)

HTA 5 – Teacher acts on a red flag (enlarged, part 2 of 3)



HTA 5, part 2 of 3 (read left to right; parts overlap slightly)

HTA 5 – Teacher acts on a red flag (enlarged, part 3 of 3)



HTA 5, part 3 of 3 (read left to right; parts overlap slightly)

HTA 6 – Parent follows a child's progress (bonus)

Parent view

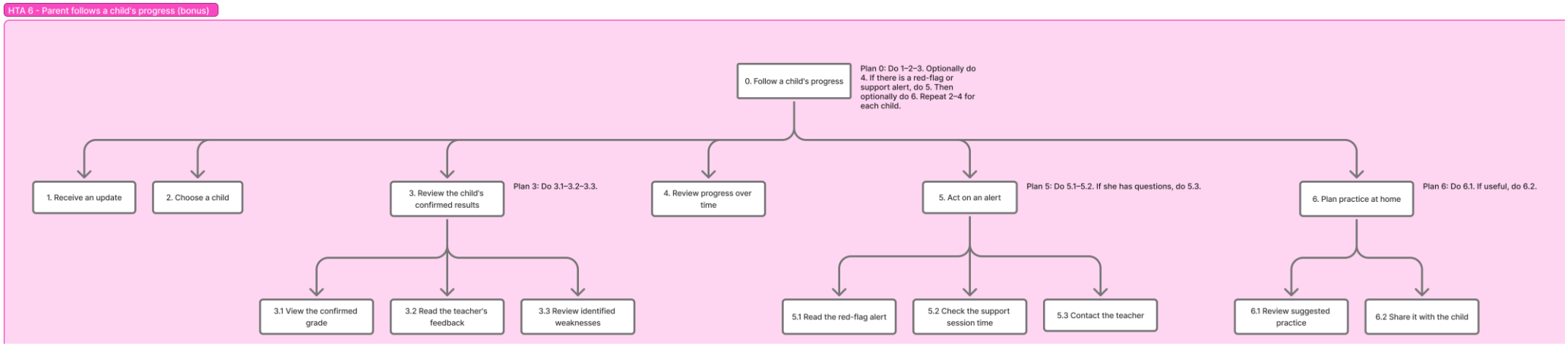
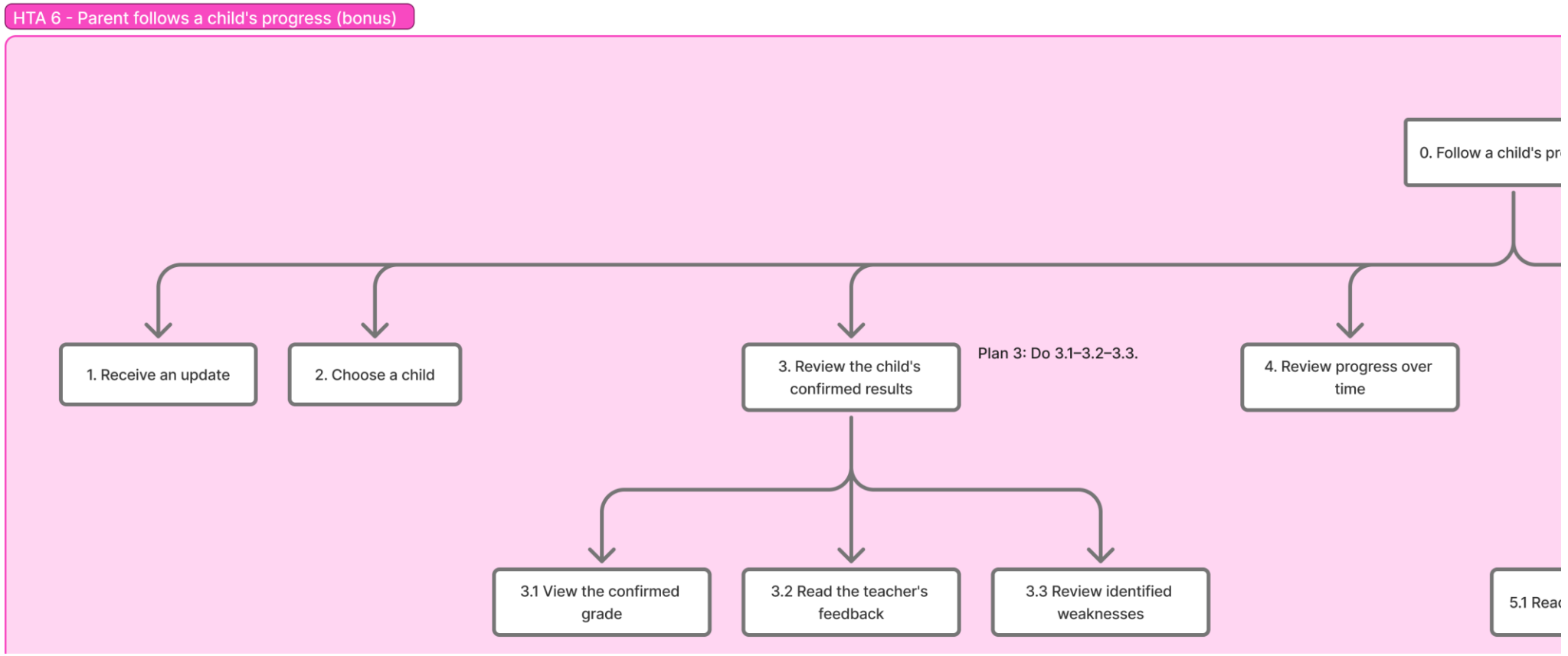


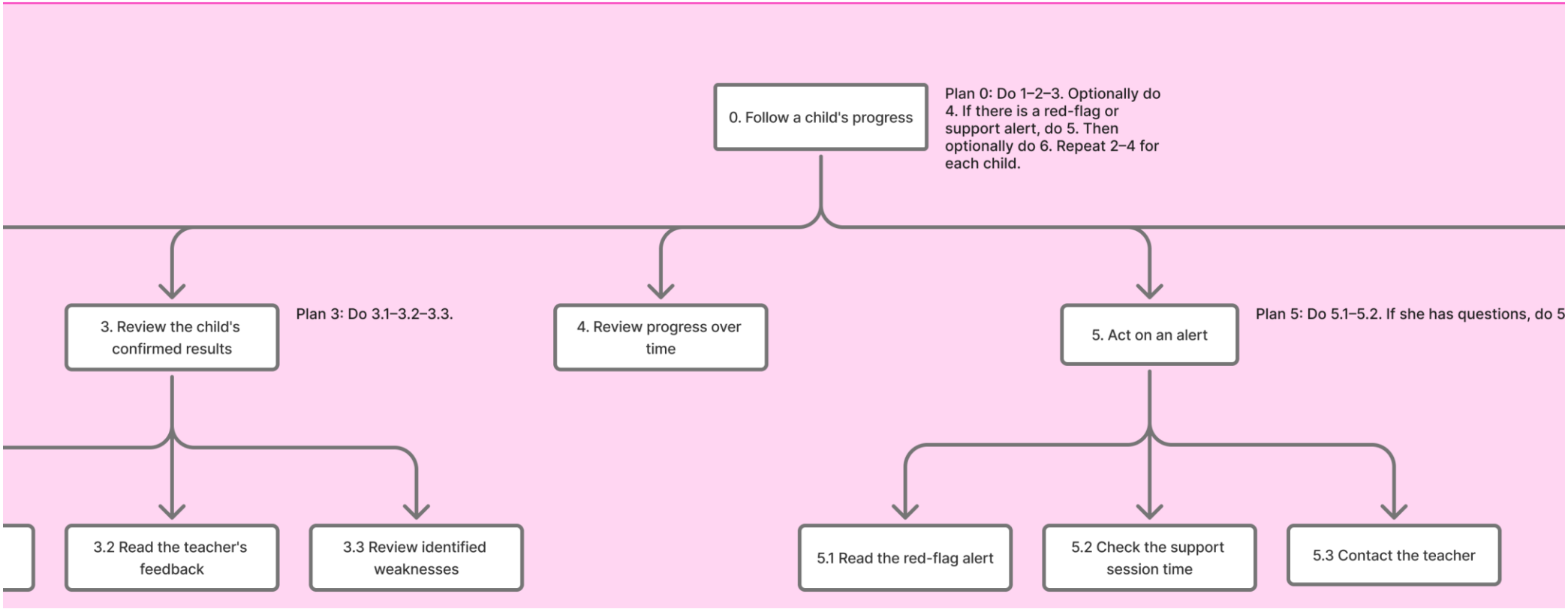
Figure 6.0 – HTA 6 in full

HTA 6 – Parent follows a child's progress (bonus) (enlarged, part 1 of 3)



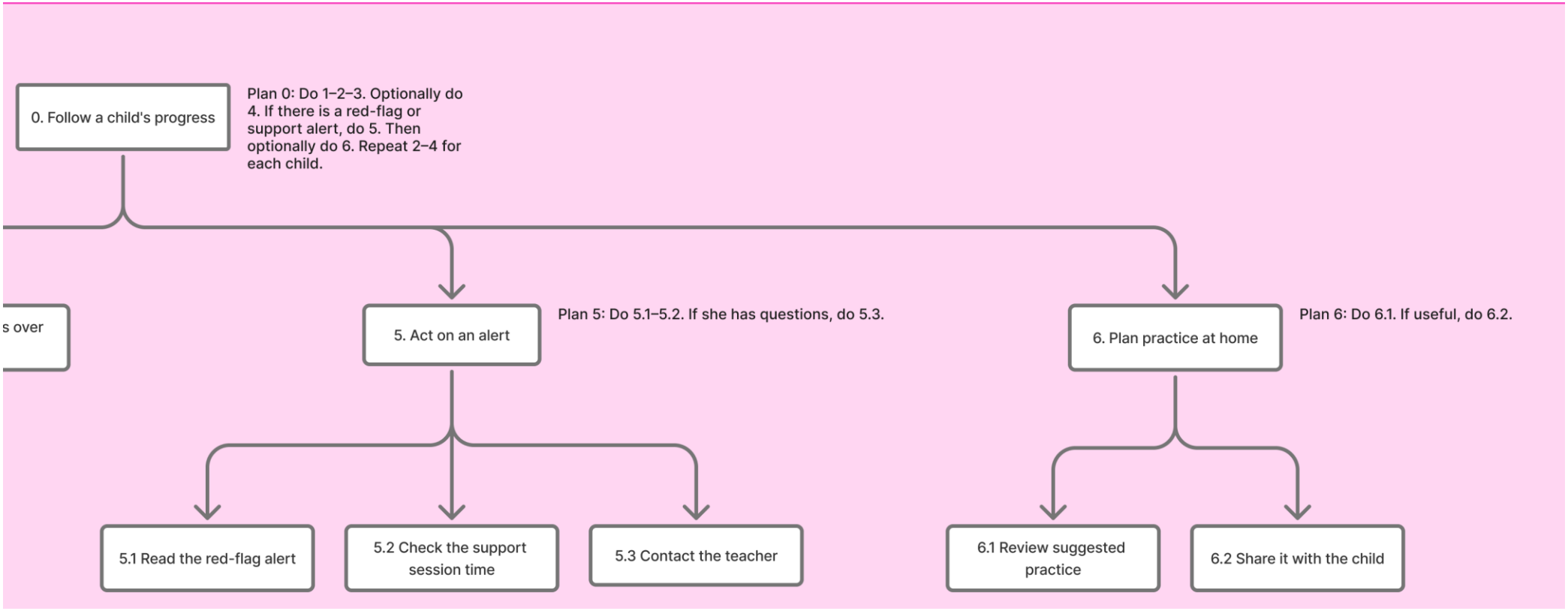
HTA 6, part 1 of 3 (read left to right; parts overlap slightly)

HTA 6 – Parent follows a child's progress (bonus) (enlarged, part 2 of 3)



HTA 6, part 2 of 3 (read left to right; parts overlap slightly)

HTA 6 – Parent follows a child's progress (bonus) (enlarged, part 3 of 3)



HTA 6, part 3 of 3 (read left to right; parts overlap slightly)

2. User Personas

We created four personas using one shared template. The teacher persona is grounded in the teacher interview. The interview did not cover young children or parents, so the student and parent personas are based on team assumptions, as noted at the bottom of each card; where a detail does come from the interview (for example, parents receiving only "X/Y"), it is tied back to it.

- **Ms. Mariam Adel** – mathematics teacher (primary user of the teacher view)
- **Leila Khaled** – Grade 2 student who needs visual, encouraging feedback
- **Ali Khaled** – Grade 5 student who reads feedback on his own and wants to know what to fix
- **Dina Mostafa** – parent of Ali and Leila (bonus parent view)

Figma file: LearnLoop – Personas



Ms. Mariam Adel, 29

Mathematics Teacher · Grade 4
Cairo, Egypt

"Just tell me what's still unchecked, and stop making me type every grade into a sheet."

Background

- Gives weekly quizzes, on paper and on Google Classroom
- Marks every paper by hand against a marking scheme; online papers in a PDF editor
- Types grades into Excel, then sends parents an "X/Y" report
- Keeps a personal notes document of each student's problem areas, updated weekly

Scenario

After collecting this week's quizzes, Mariam marks each paper while flipping back and forth to the marking scheme. She reaches an answer she can't read and sets the paper aside to ask a colleague. Once marking is done, she types every grade into Excel, sends parents their "X/Y" report, and updates her notes document. She notices one student has scored low two weeks in a row, so she messages him to book an office-hours session. Results reach students 2–4 days after the quiz.

Tech Profile

Tech Comfort ●●●●○

Devices: Laptop, using Excel, a PDF editor and Google Classroom

Goals

- Return results faster, as instant notifications
- See what percentage is still unchecked or ungraded
- Have grades and progress recorded automatically
- Spot students with consistent mistakes early

Behaviours

- Checks everything against the marking scheme, and would do the same with AI results
- Asks a colleague about unreadable answers; if no one can read them, counts them as unanswered
- Reaches out after two weeks of low marks and offers online office hours
- Prefers a light, clean, simple layout

Frustrations

- Flipping between the paper and the marking scheme
- Typing grades into the spreadsheet by hand
- Updating the progress spreadsheet manually
- Students who clarify handwriting later can't get the marks back

Based on teacher interview, as well as some assumptions.

Figure 2.1 – Persona - Teacher



Leila Khaled, 7

Primary Student · Grade 2
Cairo, Egypt

"Did I get a star? Is Miss happy with me?"

Background

- In Grade 2; can read short, simple words but not long sentences
- Recognises pictures, colours and icons more easily than text
- Uses the family iPad; a parent helps her take and upload the photo of her exam

Scenario

Leila's parent takes a photo of her maths quiz on the iPad and helps her send it. A few days later Leila opens the app and sees a big smiley face and three stars. She taps the star to hear "Great job with adding!" and sees a small picture of a clock showing she needs more practice telling time.

Tech Profile

Tech Comfort

Devices: Family iPad

Based on team assumptions; the teacher interview did not cover young children.

Goals

- Know quickly if she did well
- Feel proud of her work
- Know what to practise next, in a simple way

Needs

- Visual feedback
- Big buttons
- Encouragement
- No comparison with classmates

Frustrations

- Marks X/Y do not make sense to her
- Waiting for days does not suit her
- Long feedback is confusing

Figure 2.2 – Persona - Student Grade 2



Ali Khaled, 10

Primary Student · Grade 5
Cairo, Egypt

“If there's just a cross, I don't know what I did wrong. I have to guess.”

Background

- Leila’s older brother
- In Grade 5; reads and understands written feedback on his own
- Finds fractions hard, especially comparing them
- Uses the family iPad and his dad's laptop without help
- Prefers asking a friend or his dad over asking the teacher in class

Scenario

Ali gets his maths quiz back with 6/10 and two crosses next to the fraction questions. He doesn't know what he did wrong, and he doesn't want to ask in front of the class. One answer he's sure was right, but the teacher read his 8 as a 3. He tells her afterwards, but the marks are already final. At home he tries to work it out with his dad, but neither of them knows exactly what to practise.

Tech Profile

Tech Comfort ●●●○○

Devices: Family iPad, father’s laptop

Goals

- Get better at fractions
- Know exactly what to practise
- Get help without feeling embarrassed
- See his progress over time

Needs

- Badges and streaks
- Private progress tracking
- A “Need Help” button
- Clear Weakness Explanation
- Points for effort and a personal best to beat

Frustrations

- A cross with no explanation
- Lost marks for handwriting even after clarifying
- Embarrassed to ask in class

Based on team assumptions; the teacher interview did not cover young children.

Figure 2.3 – Persona - Student Grade 5



Dina Mostafa, 38

Parent of Ali (Grade 5) & Leila (Grade 2)
Cairo, Egypt

“All I get is ‘7 out of 10’. What should we practise at home?”

Background

- Mother of Ali and Leila; works full-time
- Helps Leila take photos and upload her exams
- Gets a weekly report: “This week’s quiz mark is X/Y”
- Checks the children’s schoolwork in the evenings

Scenario

On Thursday evening Dina gets the weekly report: “This week’s quiz mark for Ali is 4.5/10.” She can see the mark went down, but not why or what to practise. Only at the next parents’ meeting does she find out that Ali has struggled with fractions for weeks.

Tech Profile

Tech Comfort ●●●○○

Devices: Smartphone, family iPad

Goals

- Know which topics each child finds hard
- Know how to help at home
- Hear early when a child is falling behind
- Keep track of two children without extra effort

Needs

- Confirmed weaknesses, not just a mark
- A short progress summary per child
- Alerts for red flags and booked support sessions
- Everything readable on her phone in a few minutes

Frustrations

- Gets only a mark, with no feedback or weaknesses
- Finds out about problems late
- Doesn’t know what to practise with her children

Based on the teacher interview; all other details are team assumptions.

Figure 2.4 – Persona - Parent

3. Features (Functional Requirements)

This table lists every functionality the LearnLoop interface will include, grouped by where each feature comes from: the course brief, our unique feature (CGV), the teacher interview, our personas, and features added after Milestone 1. F-IDs are fixed and are used to label features in the HTAs and wireframes.

Legend

- **Student view / Teacher view** – features required by the course brief.
- **CGV – Confidence-Guided Verification** – LearnLoop's unique feature from Milestone 1.
- **Derived from the teacher interview** – features based on the interview findings.
- **Derived from personas and the M1 target audience** – features that meet the needs of our student personas.
- **Extra features** – effort points, personal best and class goal, added after Milestone 1 to motivate practice without ranking students.
- **Parent view** – the optional view for parents.
- **View column** – who uses the feature: Student, Teacher, Parent or Both.

Feature table

F-ID	Feature	Description	View
Student view			
F1	Exam upload	Student uploads a scanned exam by photographing or selecting each page, checking the image is clear before submitting.	Student
F2	View confirmed grade	Student sees their grade only after the teacher has confirmed it.	Student
F3	View feedback, weaknesses and red flags	Student sees the teacher-confirmed feedback, corrected answers, identified weaknesses and any red flag.	Student
F4	Result and teacher notifications	Student is notified as soon as results are released, and whenever the teacher sends feedback, comments or practice recommendations.	Student
F5	Progress across exams	Student follows their results and weak topics across exams over time.	Student
Teacher view			
F6	OCR text verification and correction	Teacher reviews the text extracted by OCR and corrects misread answers before grading.	Teacher
F7	Confirm or override AI grade and feedback	Teacher confirms each AI-suggested mark and feedback, or overrides it with the correct mark and a short reason.	Teacher
F8	Learning gaps and recurring mistakes review	Teacher reviews weaknesses found in the exam and errors that recur across exams, and confirms or dismisses each one.	Teacher
F9	Red flag review	The system flags a student whose score in a topic is low in two consecutive exams; the teacher confirms or dismisses each flag.	Teacher

F-ID	Feature	Description	View
F10	Send notifications and comments	Teacher sends feedback, personal comments and notifications to the student, and releases the results.	Teacher
F11	English and German exam support	Core: OCR reads answers written in English or German. Mixed-language handling: the teacher sets the expected language per exam or question, and answers in another language are flagged for review.	Teacher
CGV – Confidence-Guided Verification			
F12	Confidence-tiered review	Extracted answers are sorted into "needs a decision" (low confidence), "needs a glance" (medium) and "auto-accepted" (high). Auto-accepted answers are collapsed, not hidden, and can be expanded at any time.	Teacher
F13	Adjustable confidence threshold	Teacher raises or lowers the confidence level that decides which answers are auto-accepted.	Teacher
F14	Verification summary (trust score)	For each exam, shows the share auto-accepted vs. still unchecked and the estimated time saved, e.g. "12% checked by hand, about 8 minutes saved".	Teacher
F15	Per-student handwriting profile	Every correction the teacher makes is saved to that student's handwriting profile, so future readings of their writing improve.	Teacher
F16	Class-wide OCR report	Shows which questions, subjects or handwriting styles cause the most OCR problems across the whole class.	Teacher
F17	Confidence-based grade reliability	OCR confidence is carried forward, so each AI mark and detected weakness shows how reliable it is.	Teacher
Derived from the teacher interview			
F18	Side-by-side answer, scan and marking scheme	The student's extracted answer, the original scan and the marking scheme are shown together, so the teacher no longer flips between paper and scheme.	Teacher
F19	Auto-recorded gradebook with Excel export	Confirmed grades are recorded automatically and can be exported to Excel, replacing manual data entry.	Teacher
F20	Per-student weakness tracker	Each student's confirmed weaknesses are tracked over time automatically, replacing the weekly notes document.	Teacher
F21	"I need help" button	Student asks the teacher for help on a topic directly from their feedback, without having to ask in class.	Student
F22	Schedule a support session	From a confirmed red flag, the teacher books an online office-hours session and the student is notified.	Both
F23	Clarification request before release	Teacher sends an unreadable answer to the student to explain; the exam's grades are released only once it is resolved, so students do not lose marks for handwriting alone.	Both
F24	Consult a colleague	Teacher shares an unreadable answer with a colleague for a second reading.	Teacher
F25	Class grading progress	Dashboard shows how much of the class's verification and marking is done and how much is left.	Teacher
F26	Class weakness overview	Dashboard shows the most common problem areas across the class.	Teacher
F27	Roster with student and parent contacts	Teacher can quickly find student and parent contact details.	Teacher
Derived from personas and the M1 target audience			

F-ID	Feature	Description	View
F28	Age-adapted visual feedback	Results for younger students use icons, stars, colours and short text (optionally read aloud) rather than numbers alone.	Student
F29	Gamified, private progress	Badges, streaks and effort points reward practice; no individual ranking or comparison with classmates.	Student
F30	Practice recommendations	Student receives practice activities linked to each confirmed weakness.	Student
Extra features			
F34	Effort points	Students earn points for effort: on-time upload +10, answering a clarification +5, finishing practice +5, practice streak +2 per day, asking for help +3 (once per topic per week). Grades never earn points and points are never taken away.	Student
F35	Personal best	Each week the student compares their points with their own best week, e.g. "You beat your own record by 6 points!".	Student
F36	Class goal	The teacher sets a shared class target and reward; every student's points fill one class bar, and individual contributions are never shown to classmates.	Both
Parent view			
F31	Child summary	Parent sees each child's teacher-confirmed grades, feedback and weaknesses, instead of only "This week's quiz mark is X/Y".	Parent
F32	Progress summary	Parent sees a short summary of each child's progress across exams and what to practise at home.	Parent
F33	Red flag and support alerts	Parent is alerted when the teacher confirms a red flag or books a support session, and can contact the teacher.	Parent

Change since Milestone 1

After Milestone 1 we added effort points, personal bests and a shared class goal (F34–F36) to motivate practice. Points reward effort, never grades; students compare only with their own best week; and the class works towards a goal set by the teacher, so no student is ranked against classmates, in line with our Milestone 1 target audience.

English and German exam support (F11) remains a core feature. Only mixed-language answers (both languages within one answer), listed as a gap in Milestone 1, were scaled down after the teacher interview (R11): they are handled by an expected-language setting and a review flag rather than a separate tool.

4. Low-Fidelity Wireframes

We designed 21 key screens in Figma, using a small greyscale wireframe kit (top bar, side navigation, buttons, cards, tags and placeholders). Screen sizes follow the personas:

- **Teacher screens (T1–T9):** desktop, 1440 × 1024 – Ms. Mariam works on a laptop.
- **Student screens (S1–S10):** tablet, 1024 × 768 – Leila and Ali use the family iPad; big buttons, short text and one main action per screen.
- **Parent screens (P1–P2):** phone, 390 × 844 – Dina checks progress on her smartphone.

How to read the wireframes:

- Blue pills (e.g. F12) show which feature from Section 3 each element implements.
- Numbered blue circles match the annotations printed beside each screen, explaining key interactions and the requirement (R-number) or persona need behind them.
- The caption above the annotations lists the HTA steps and features each screen supports.
- CGV verification (T3) and grading (T4) are the most detailed screens, as they carry our unique feature.

Screen flows

On the Figma canvas, solid arrows connect screens within one task and dashed arrows show hand-overs between views:

- HTA 1 (upload): S1 → S2 → S3 → S4; the submitted exam then appears for the teacher (T1).
- HTA 2 (verify): T1 → T2 → T3; if an answer is unreadable, T3 ⇄ S5 (the student clarifies before grades are released).
- HTA 3 (grade): T3 → T4 → T5 → T6; releasing results in T6 notifies the student (S10) and parent (P1).
- HTA 4 (results): S10 → S6 → S7 (ask for help) or S8 (practise); S9 shows progress, points and the class goal.
- HTA 5 (red flag): T1 → T5 → T8 / T9 (contact, schedule support, follow up).
- HTA 6 (parent): P1 ⇄ P2.

Full canvas with all flow arrows: LearnLoop – Wireframes (Figma)

Teacher screens

T1 – Dashboard – Teacher

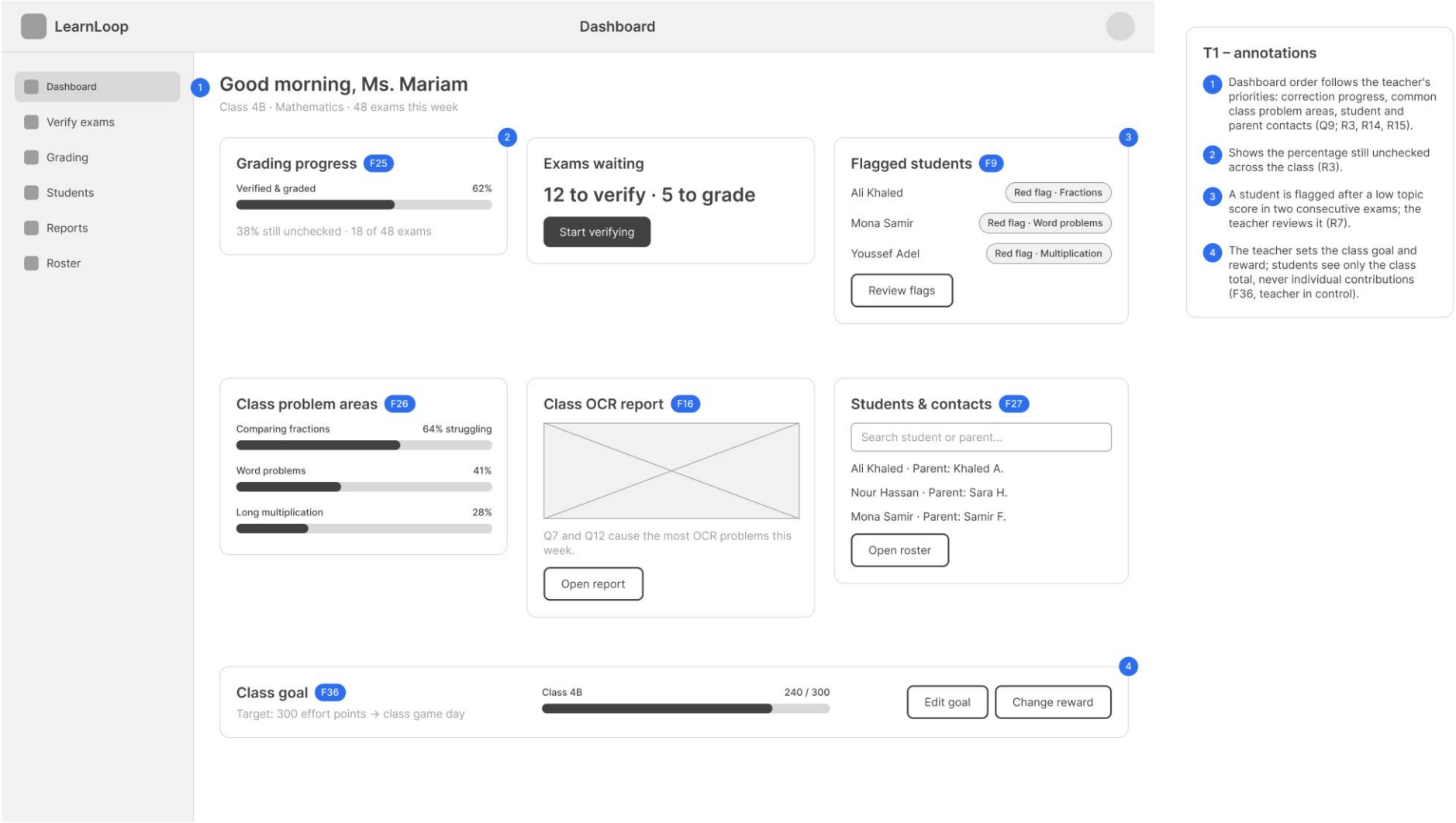


Figure 4.1 – T1 – Dashboard – Teacher

T2 – Exams to Verify – Teacher

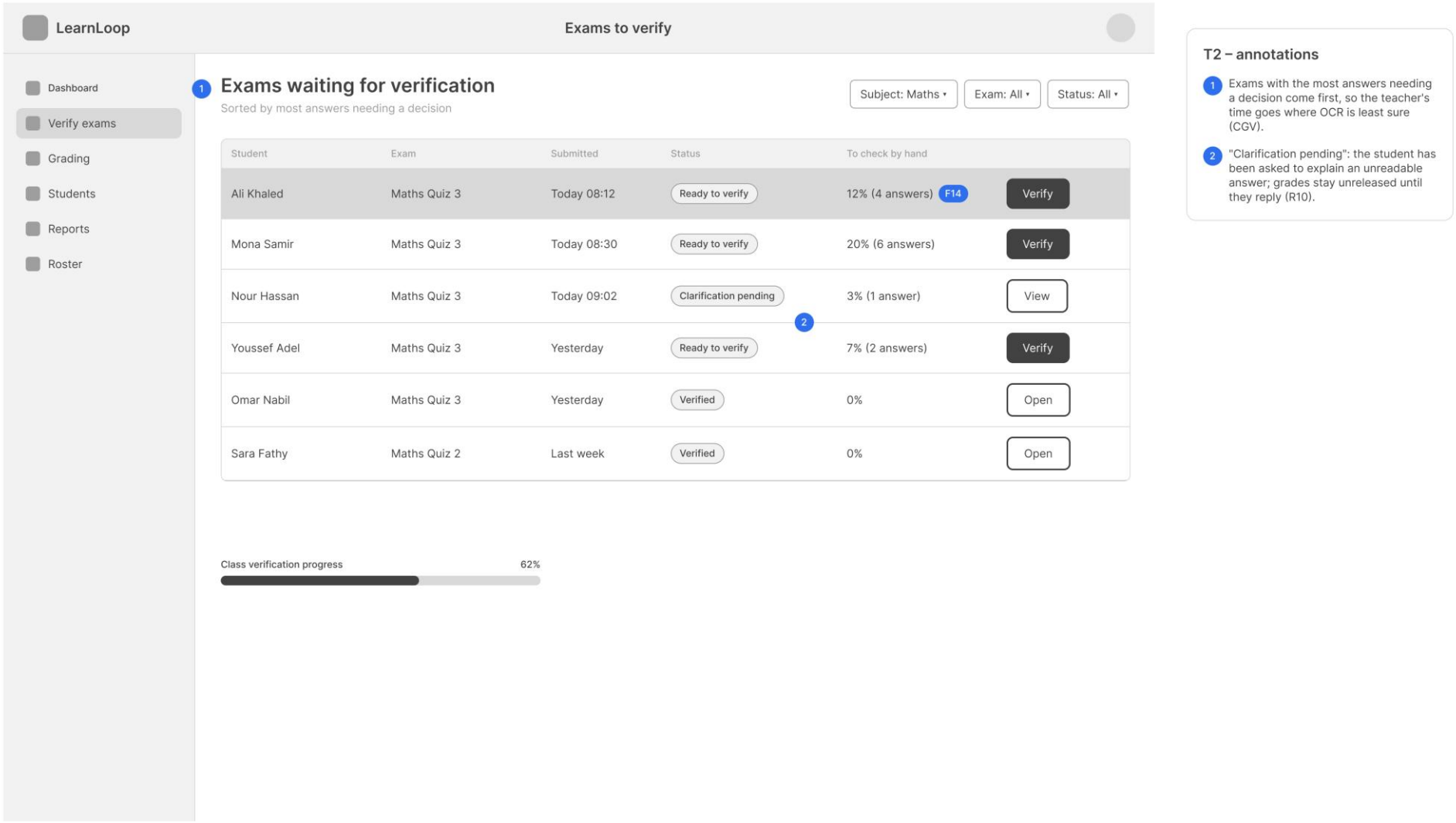


Figure 4.2 – T2 – Exams to Verify – Teacher

T3 – CGV Verification – Teacher

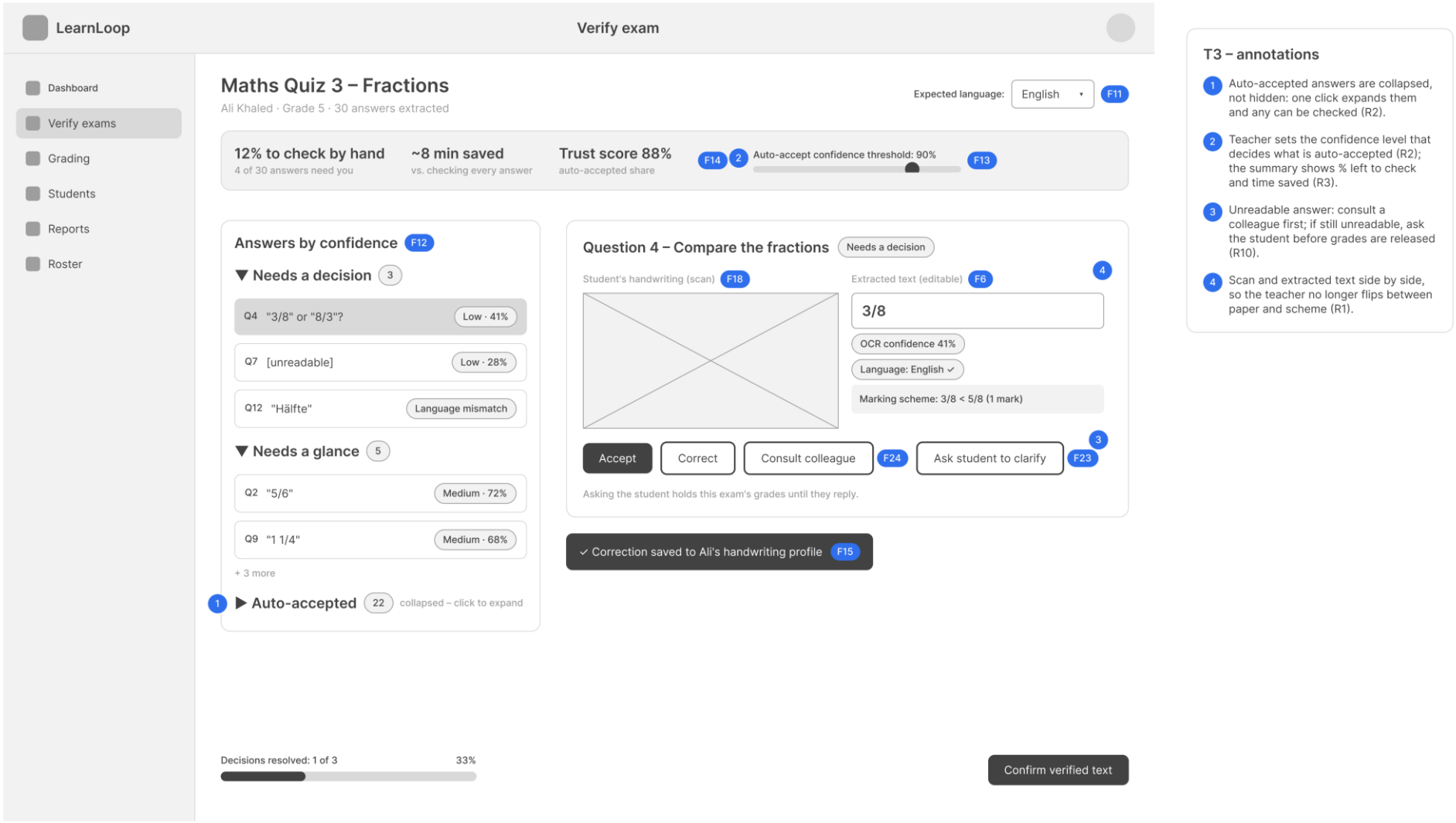


Figure 4.3 – T3 – CGV Verification – Teacher

T4 – Grading Review – Teacher

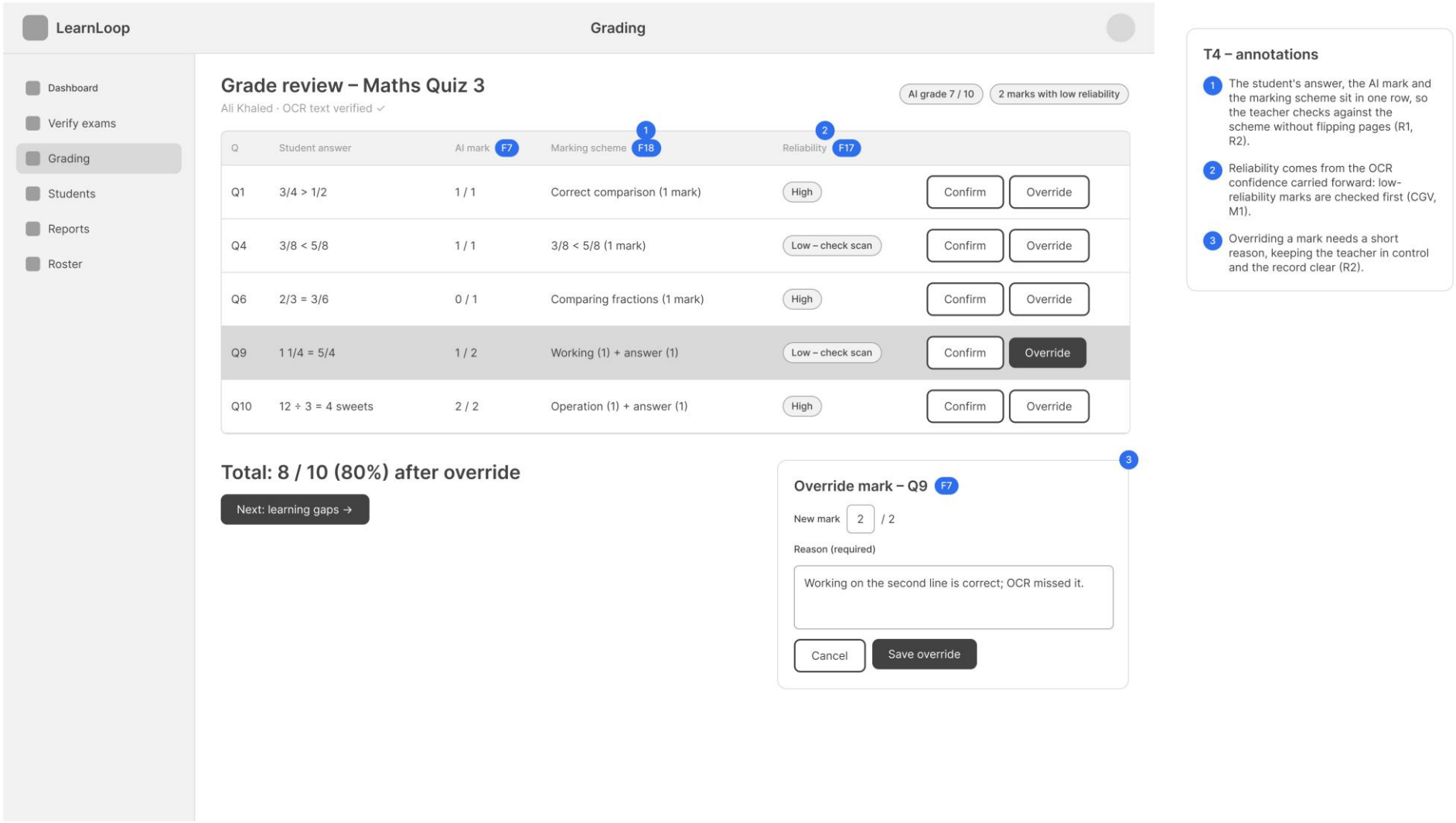


Figure 4.4 – T4 – Grading Review – Teacher

T5 – Learning Gaps & Red Flags – Teacher

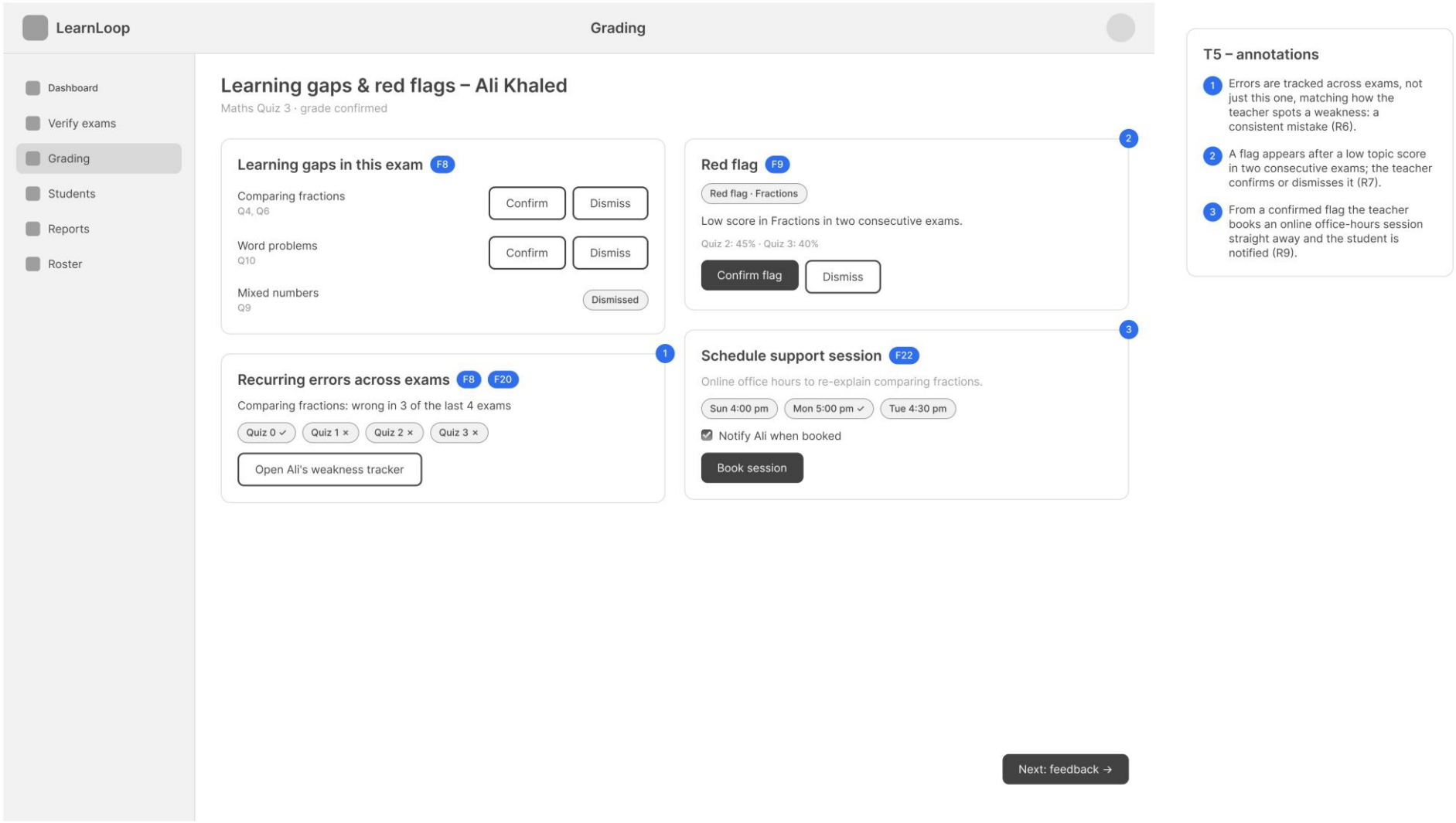


Figure 4.5 – T5 – Learning Gaps & Red Flags – Teacher

T6 – Feedback & Release – Teacher

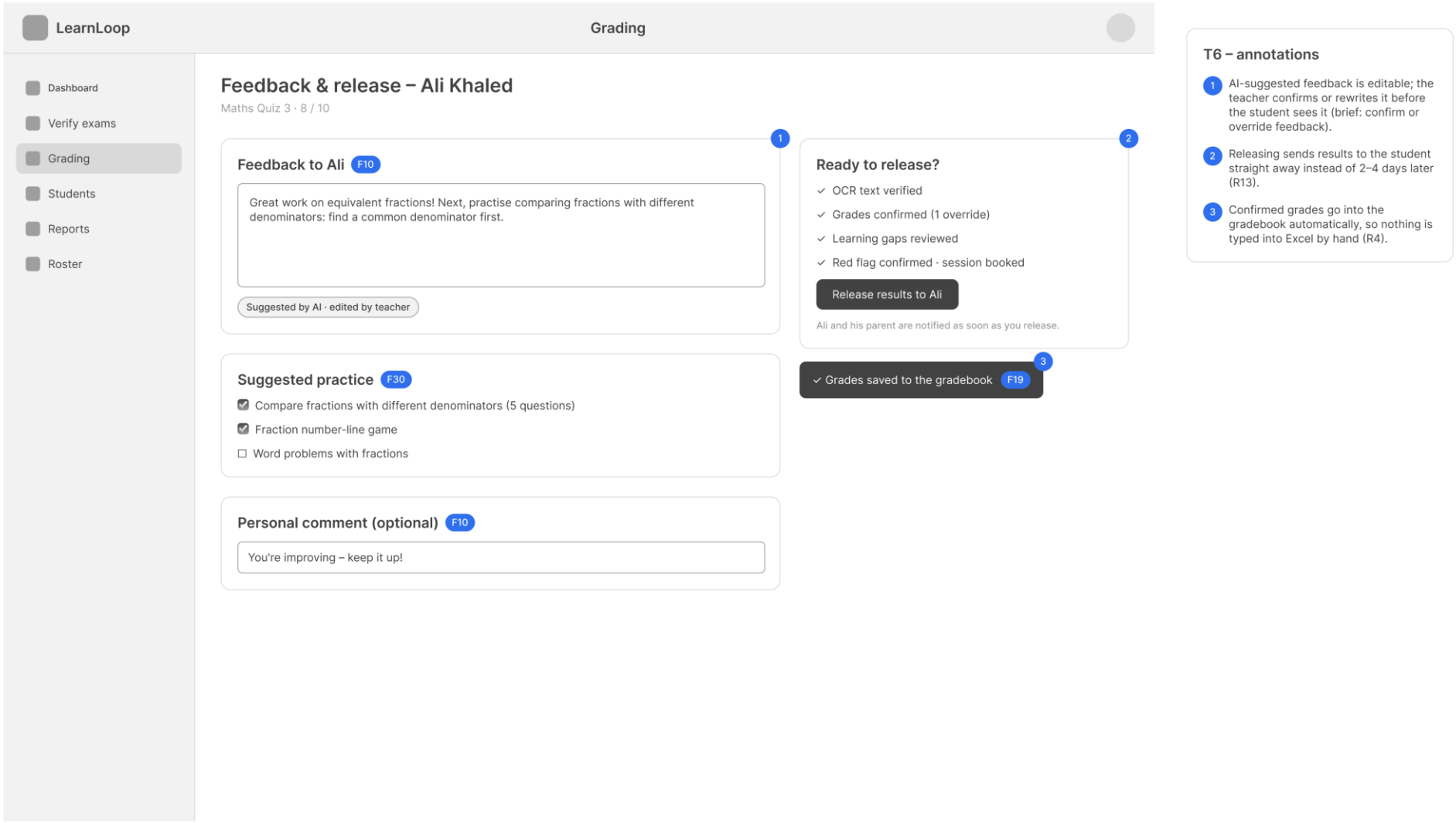


Figure 4.6 – T6 – Feedback & Release – Teacher

T7 – Class OCR Report – Teacher

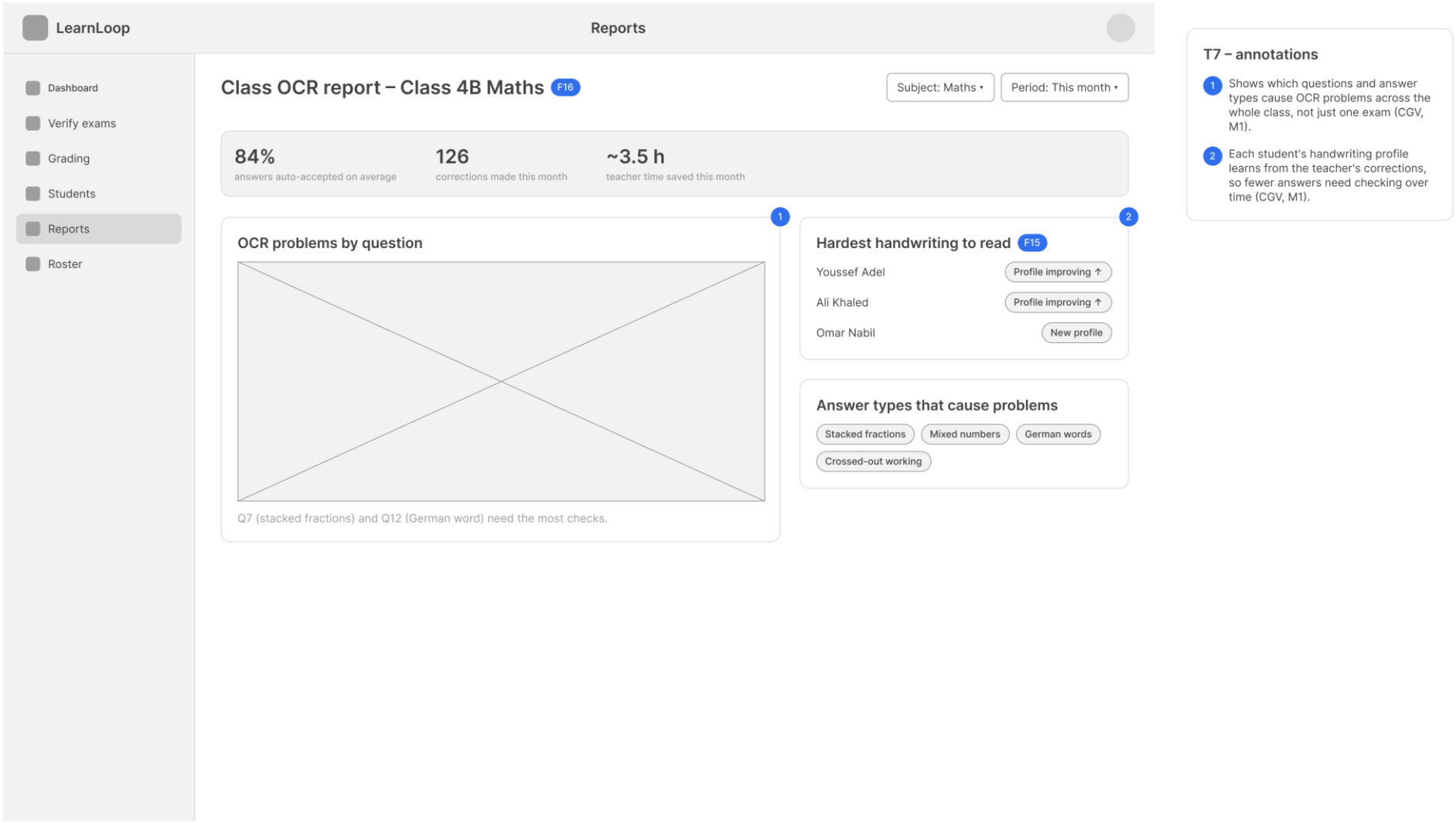


Figure 4.7 – T7 – Class OCR Report – Teacher

T8 – Roster & Gradebook – Teacher

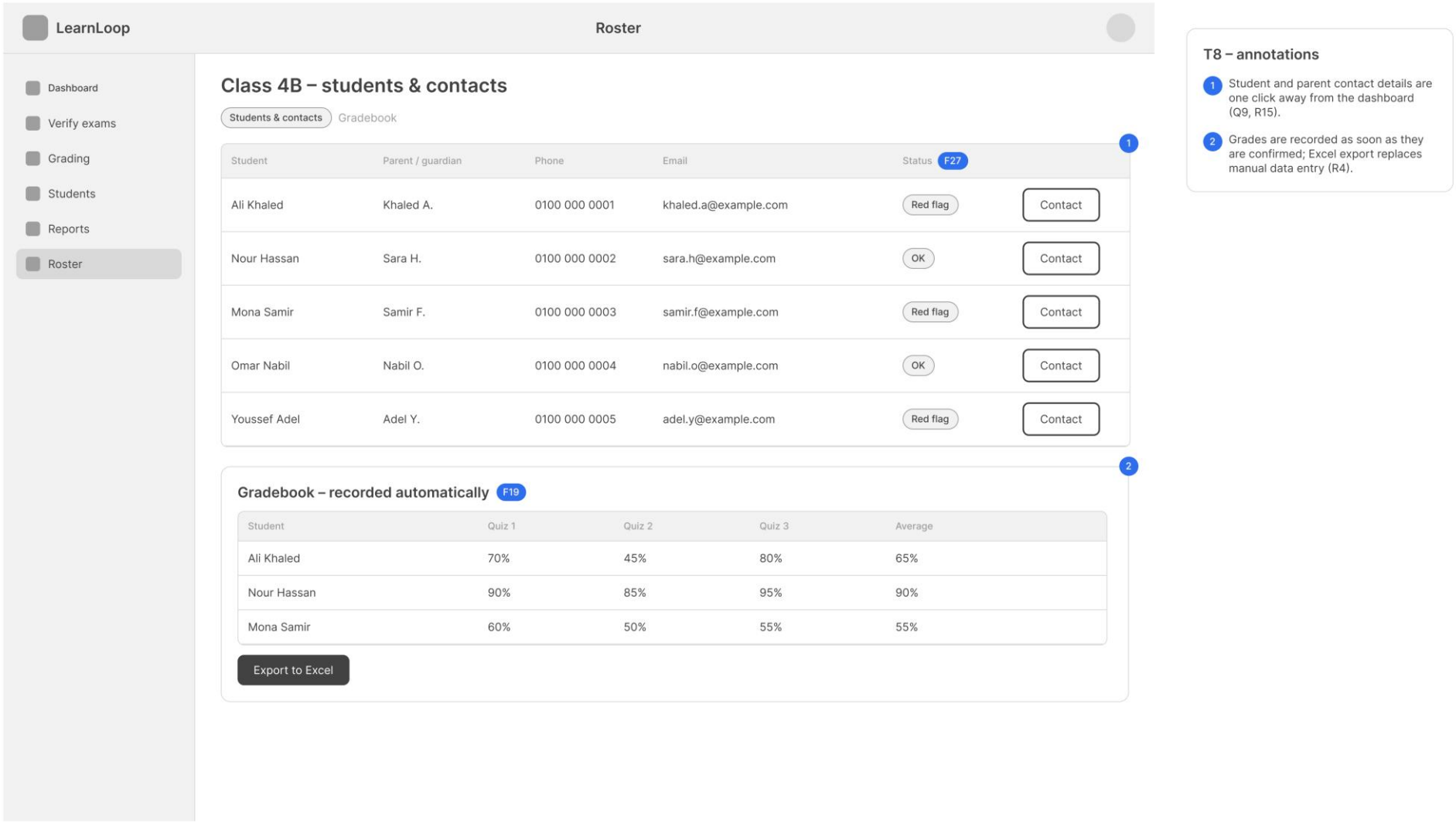


Figure 4.8 – T8 – Roster & Gradebook – Teacher

T9 – Student Profile – Teacher

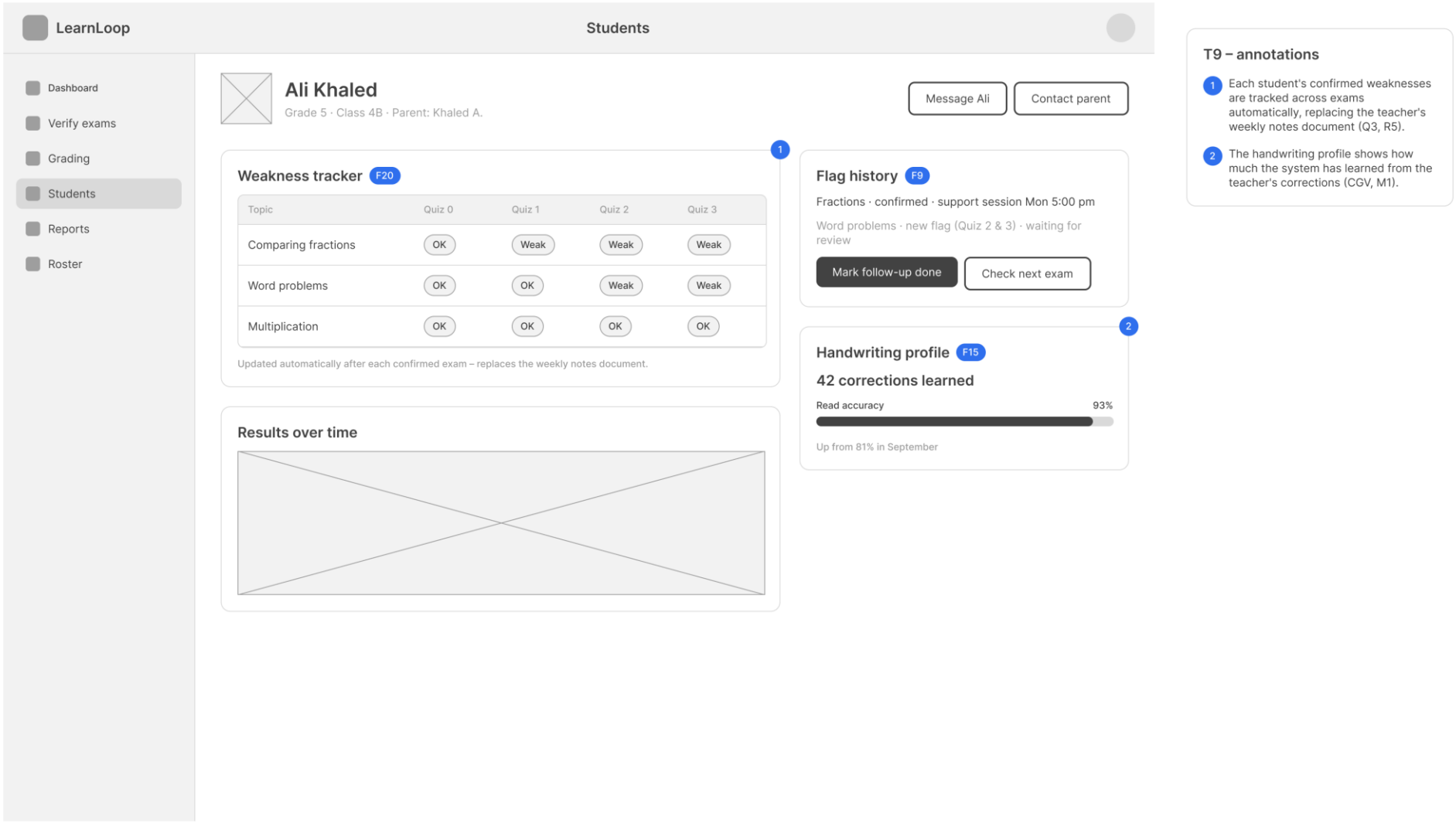


Figure 4.9 – T9 – Student Profile – Teacher

Student screens

S1 – Home – Student

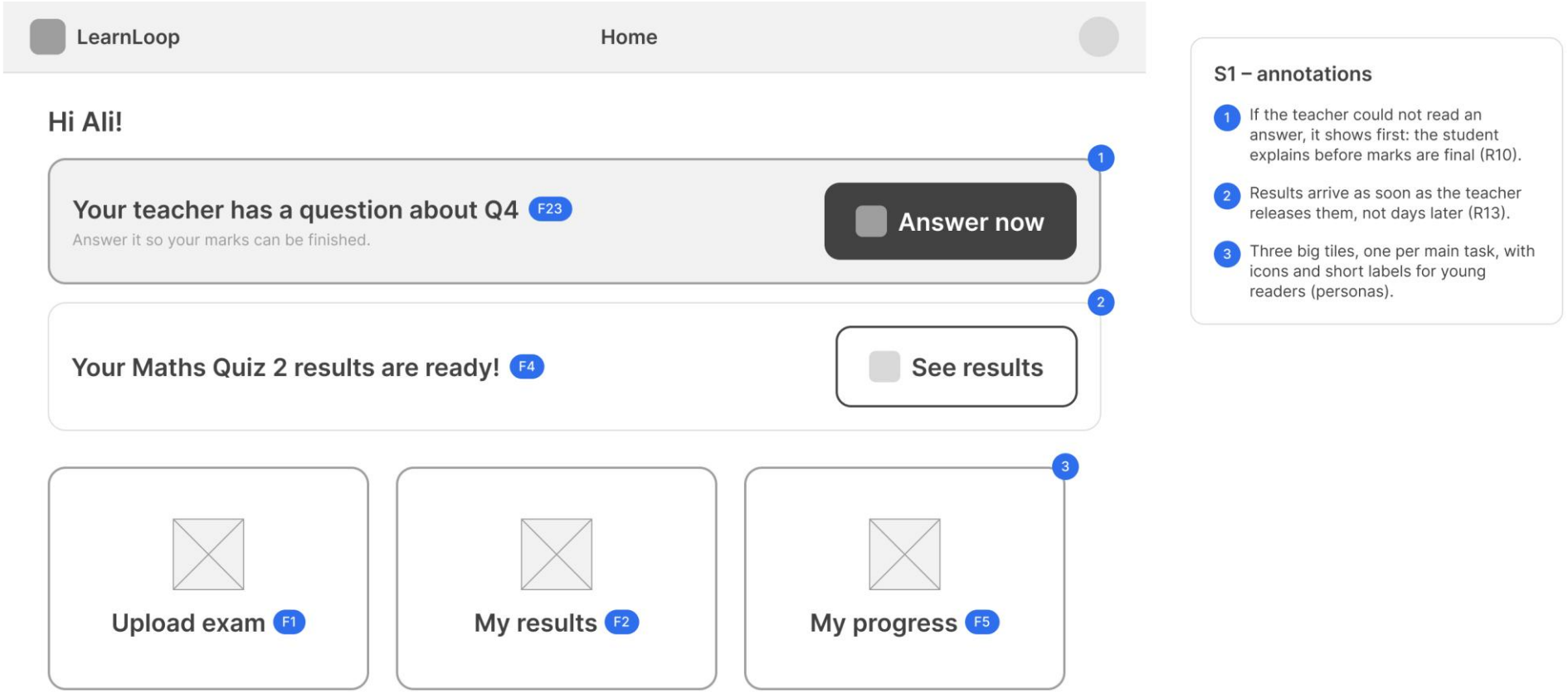


Figure 4.10 – S1 – Home – Student

S2 – Choose Exam – Student

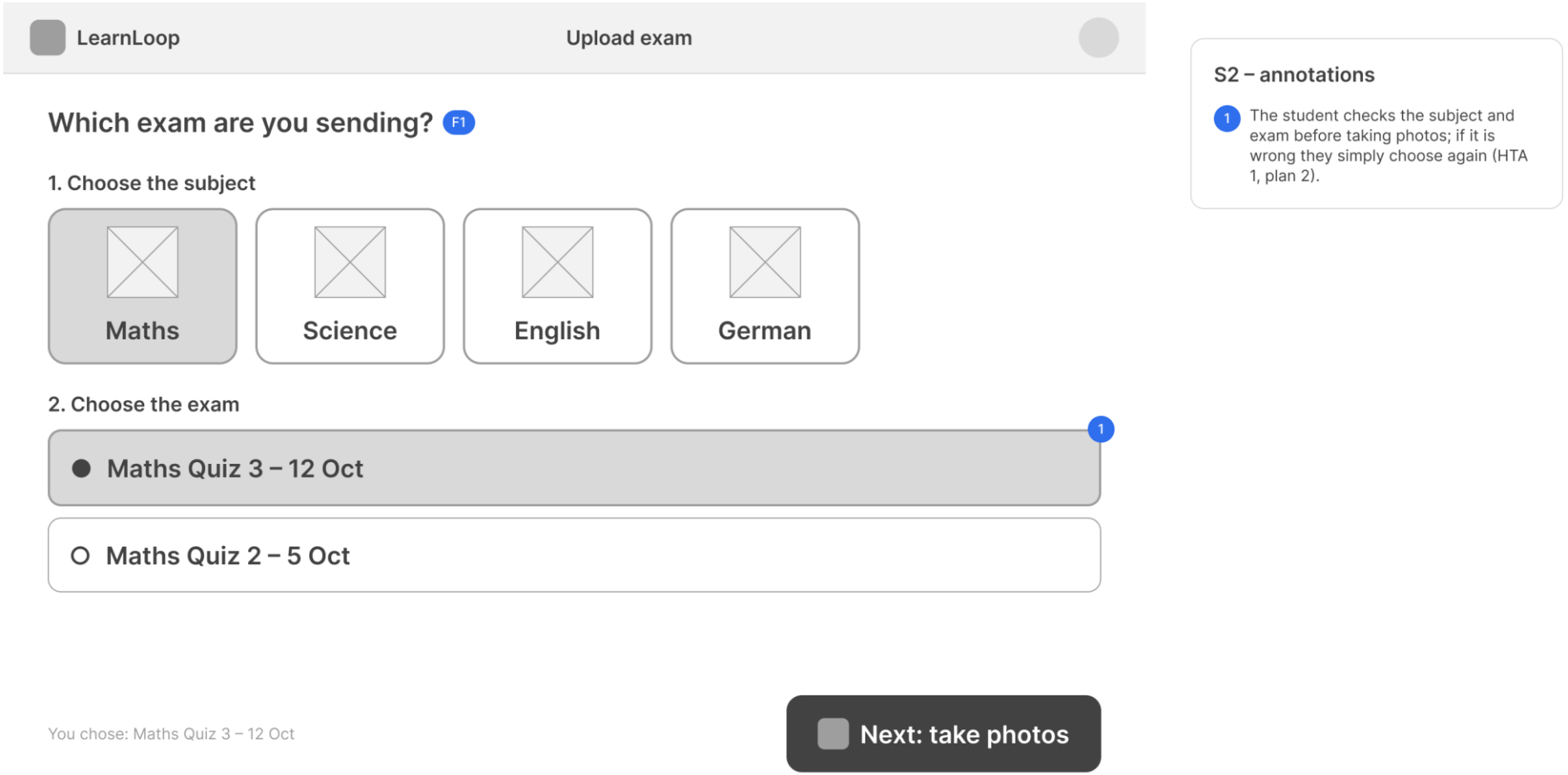


Figure 4.11 – S2 – Choose Exam – Student

S3 – Capture Pages – Student

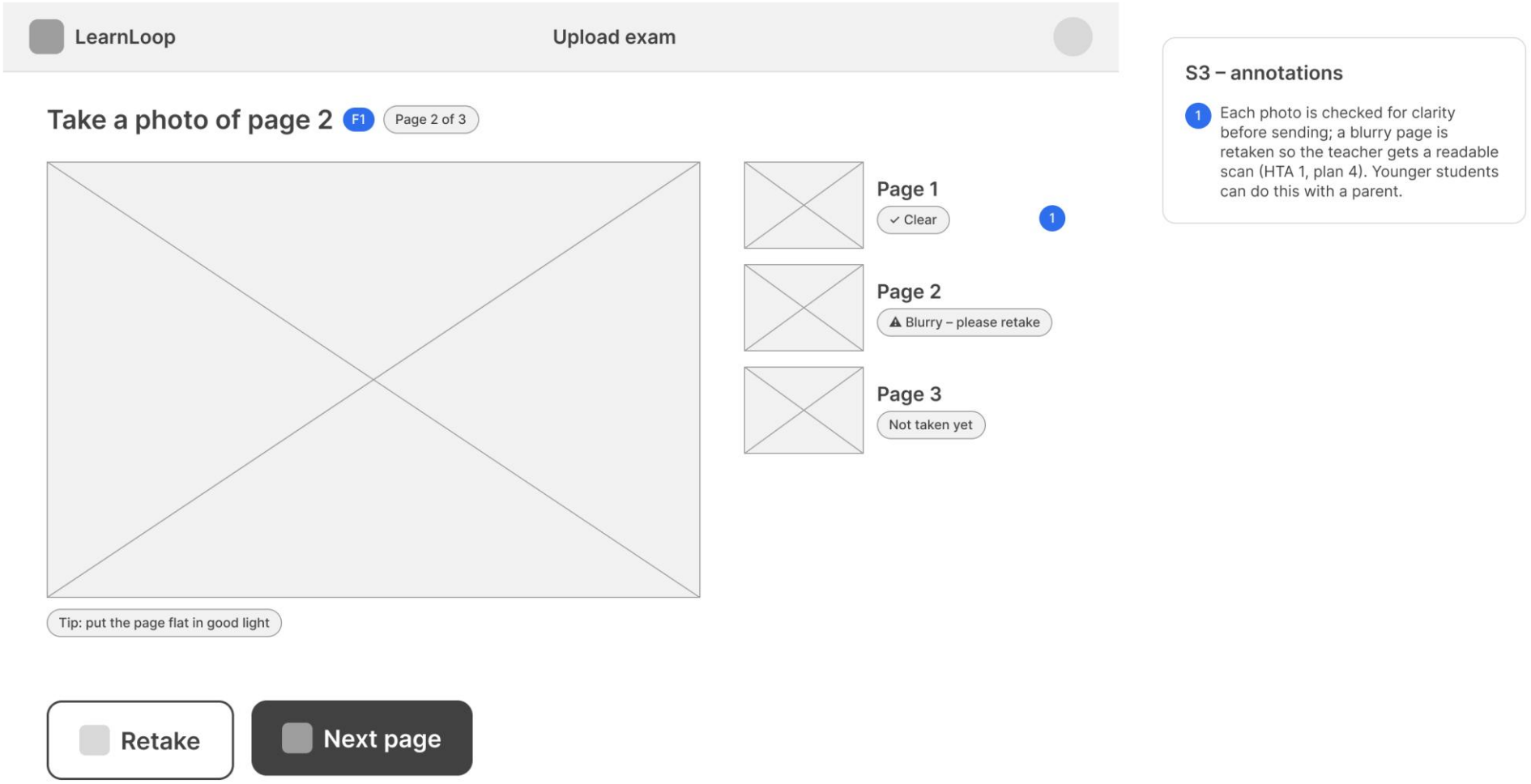


Figure 4.12 – S3 – Capture Pages – Student

S4 – Review & Sent – Student

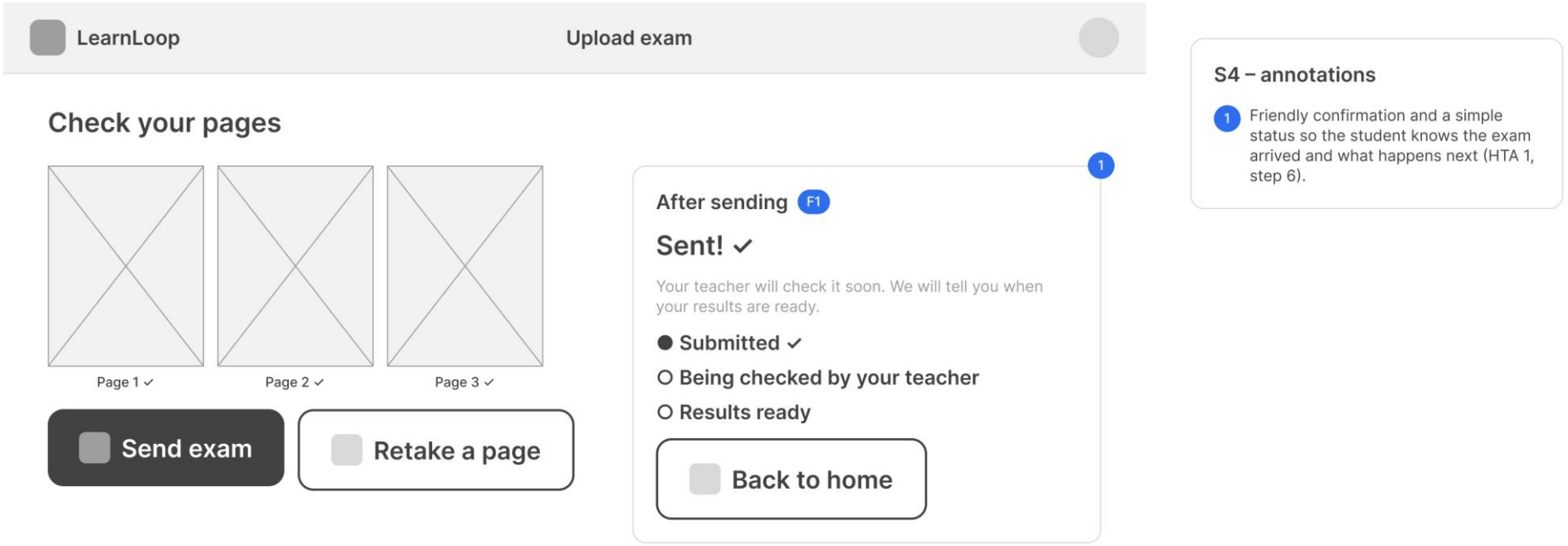


Figure 4.13 – S4 – Review & Sent – Student

S5 – Clarification Request – Student

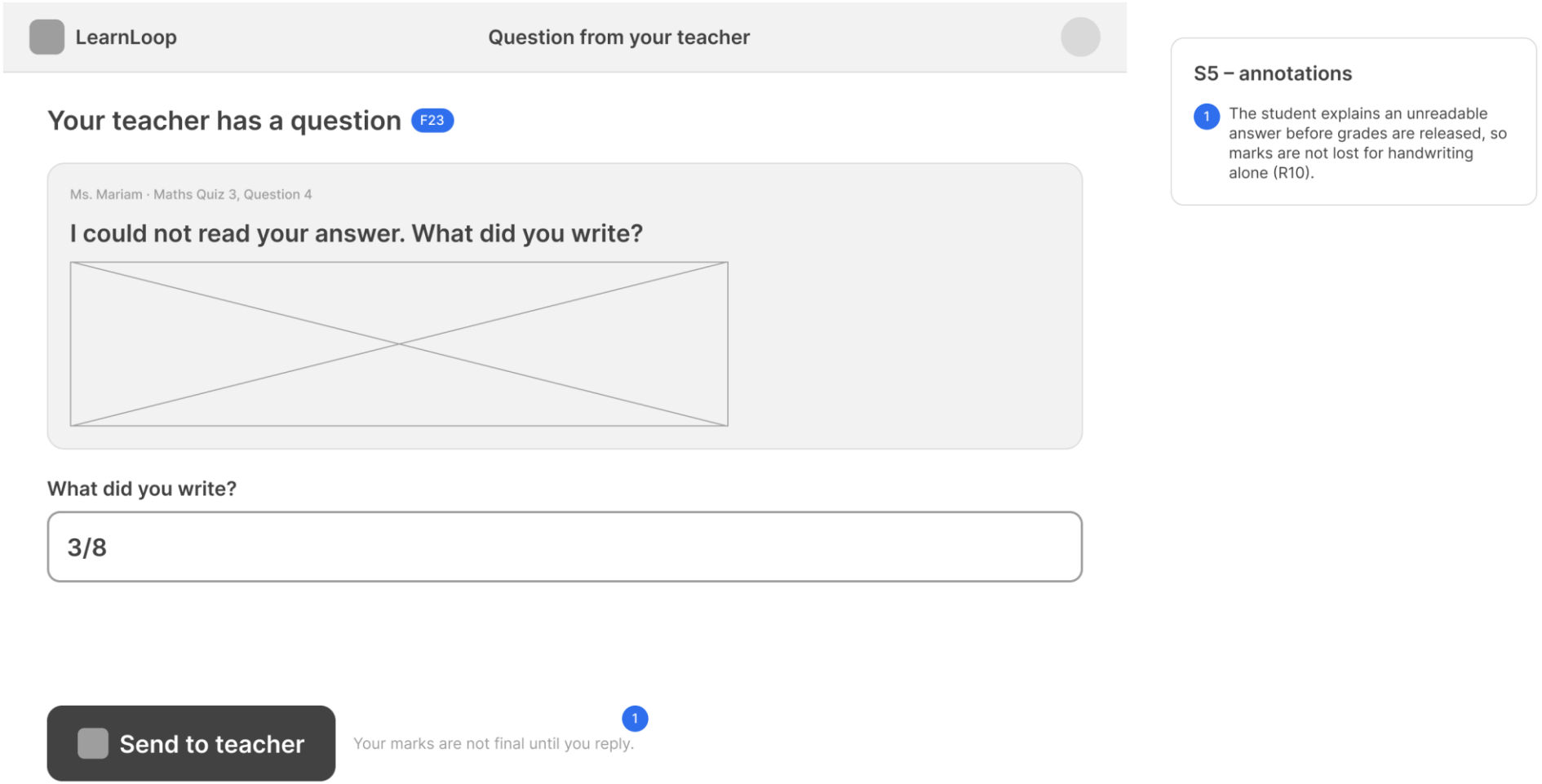


Figure 4.14 – S5 – Clarification Request – Student

S6 – Results & Feedback – Student

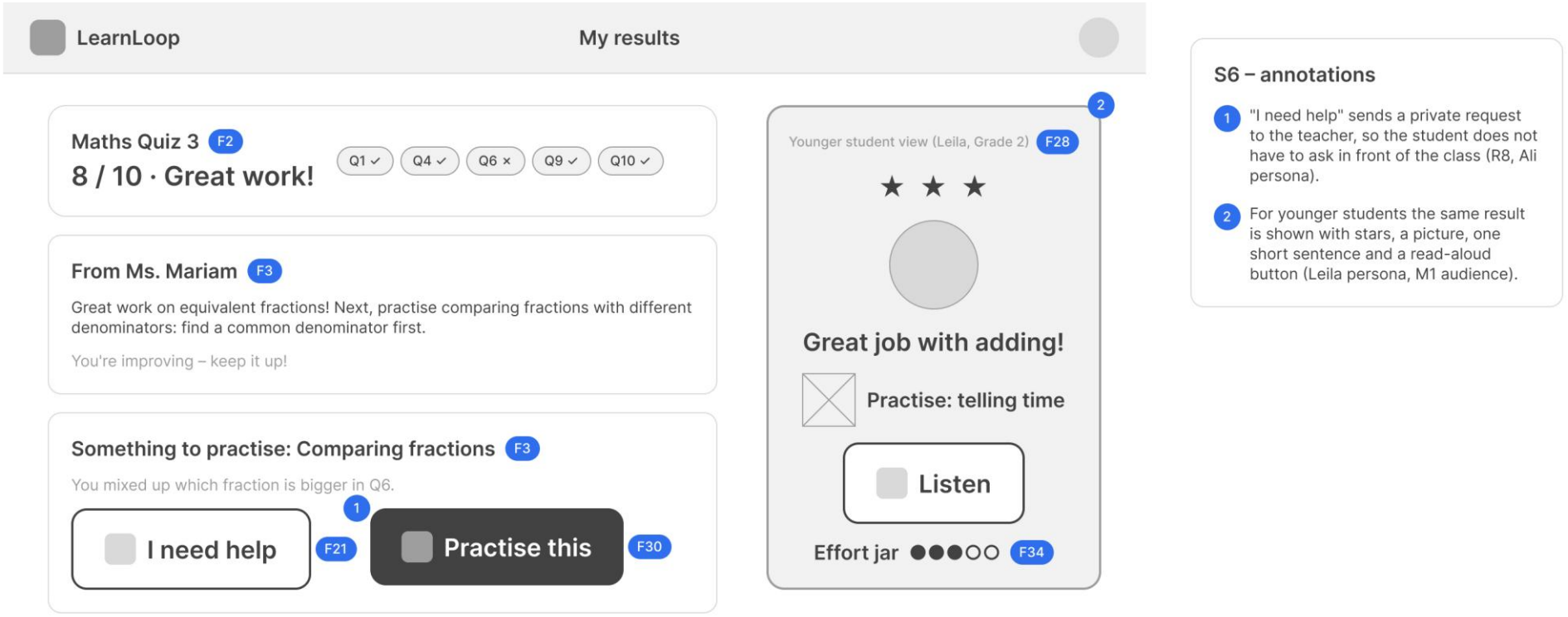


Figure 4.15 – S6 – Results & Feedback – Student

S7 – Ask for Help – Student

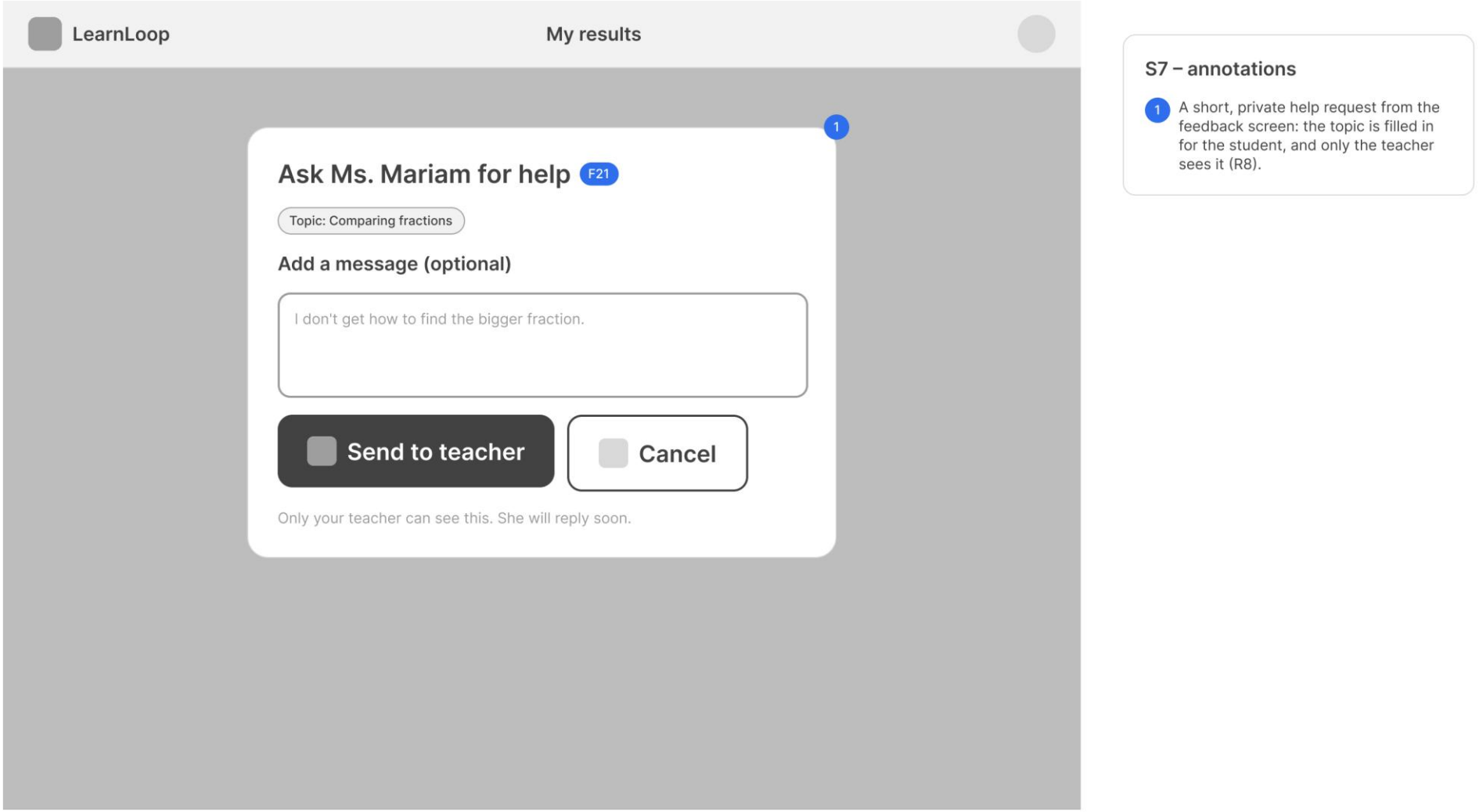


Figure 4.16 – S7 – Ask for Help – Student

S8 – Practice – Student

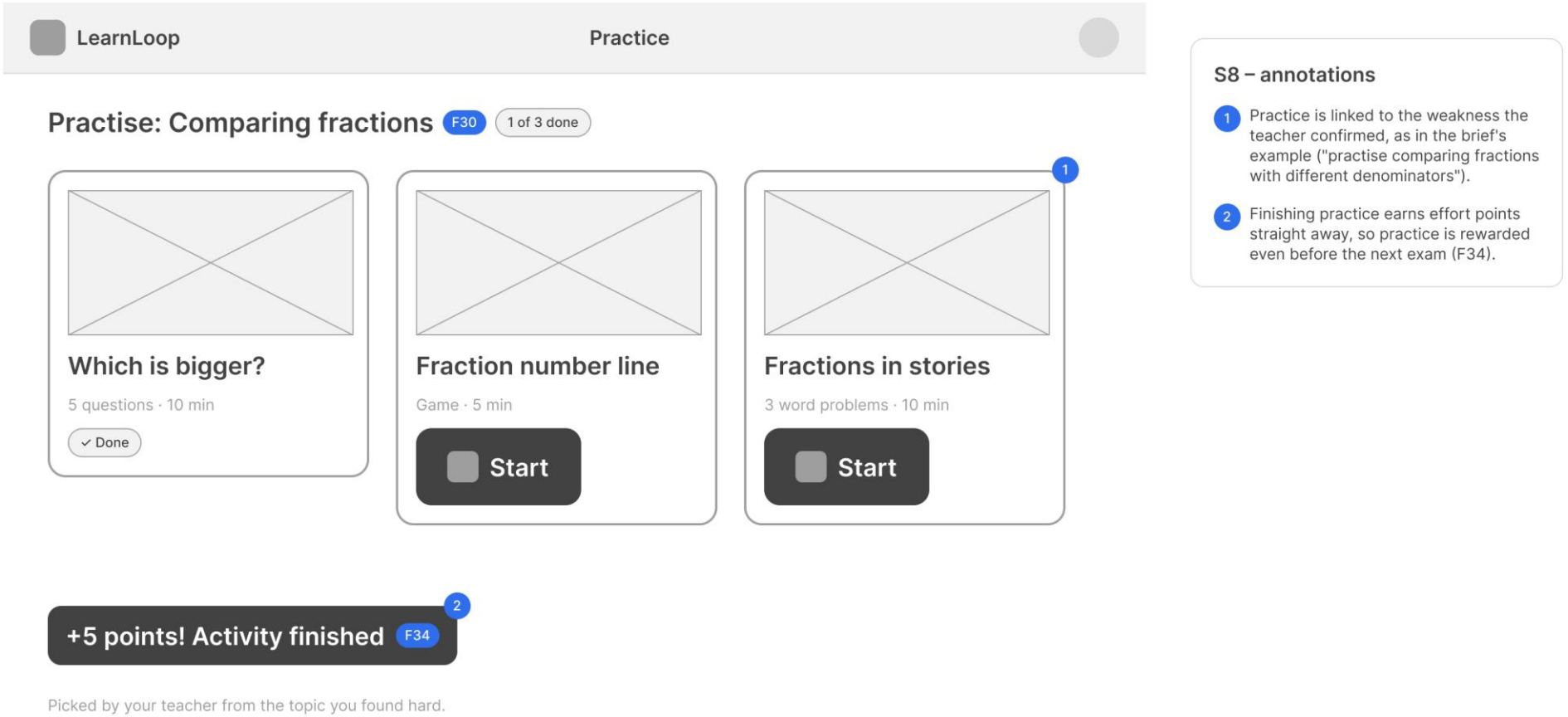


Figure 4.17 – S8 – Practice – Student

S9 – My Progress – Student

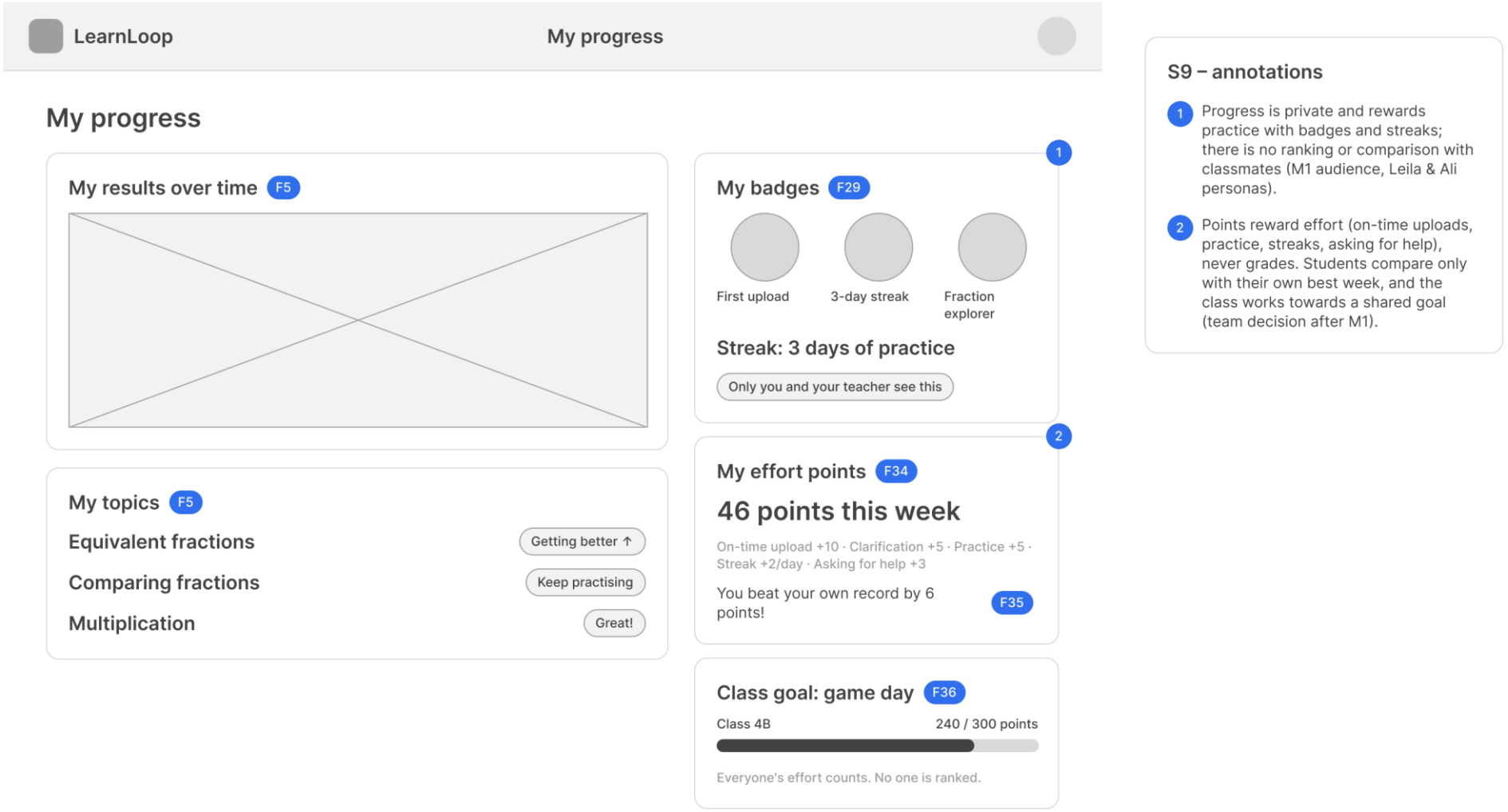


Figure 4.18 – S9 – My Progress – Student

S10 – Notifications – Student

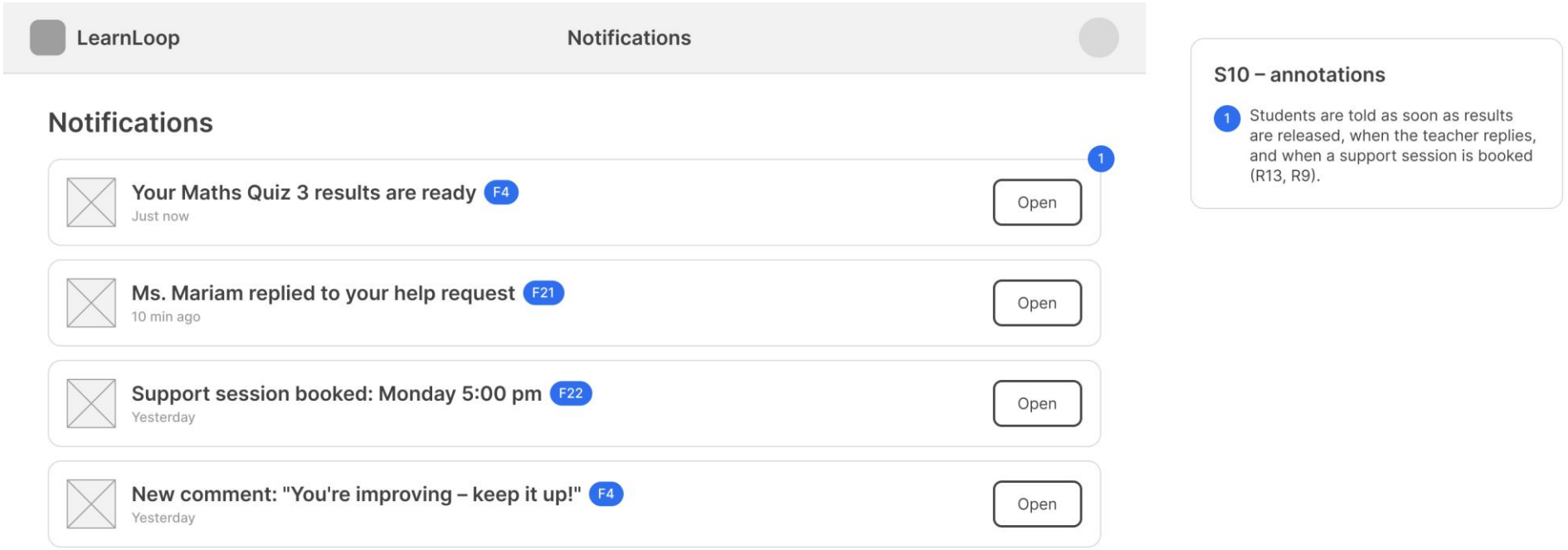


Figure 4.19 – S10 – Notifications – Student

Parent screens

P1 – Child Summary – Parent

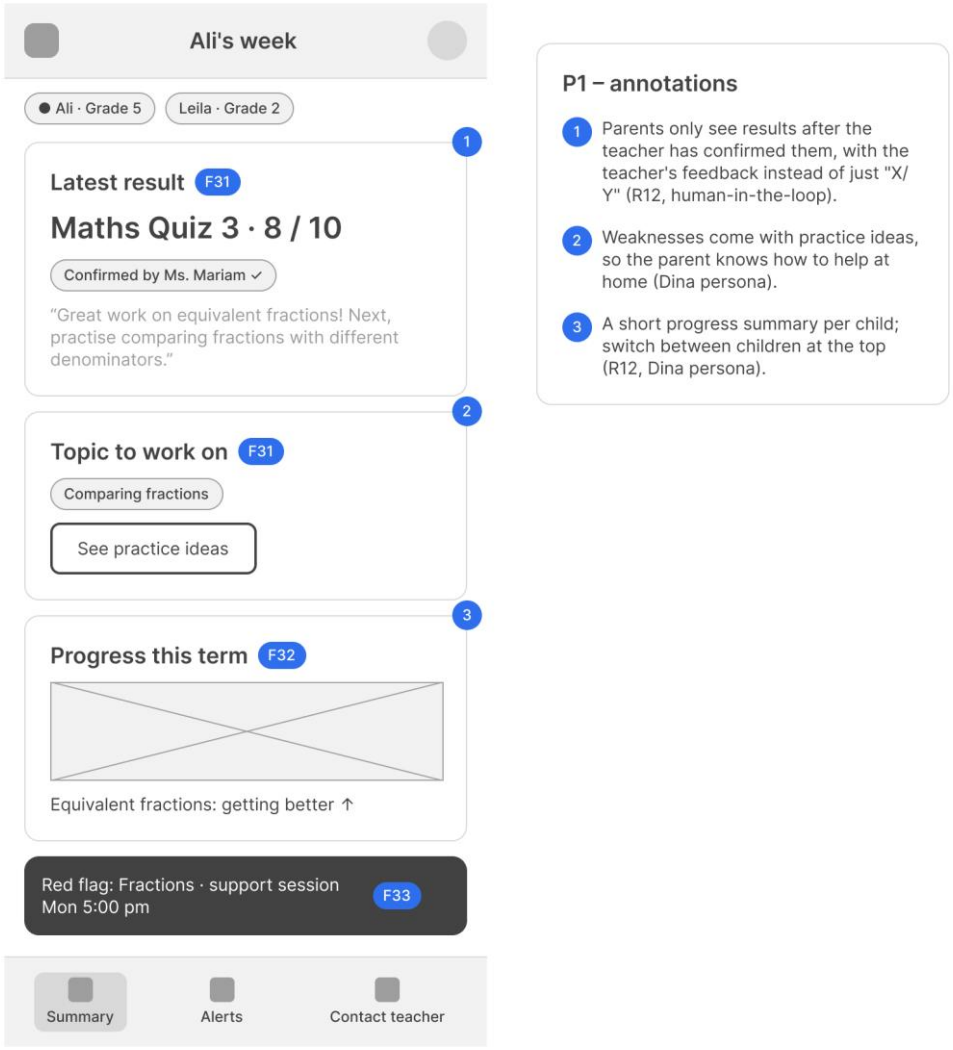


Figure 4.20 – P1 – Child Summary – Parent

P2 – Alerts – Parent

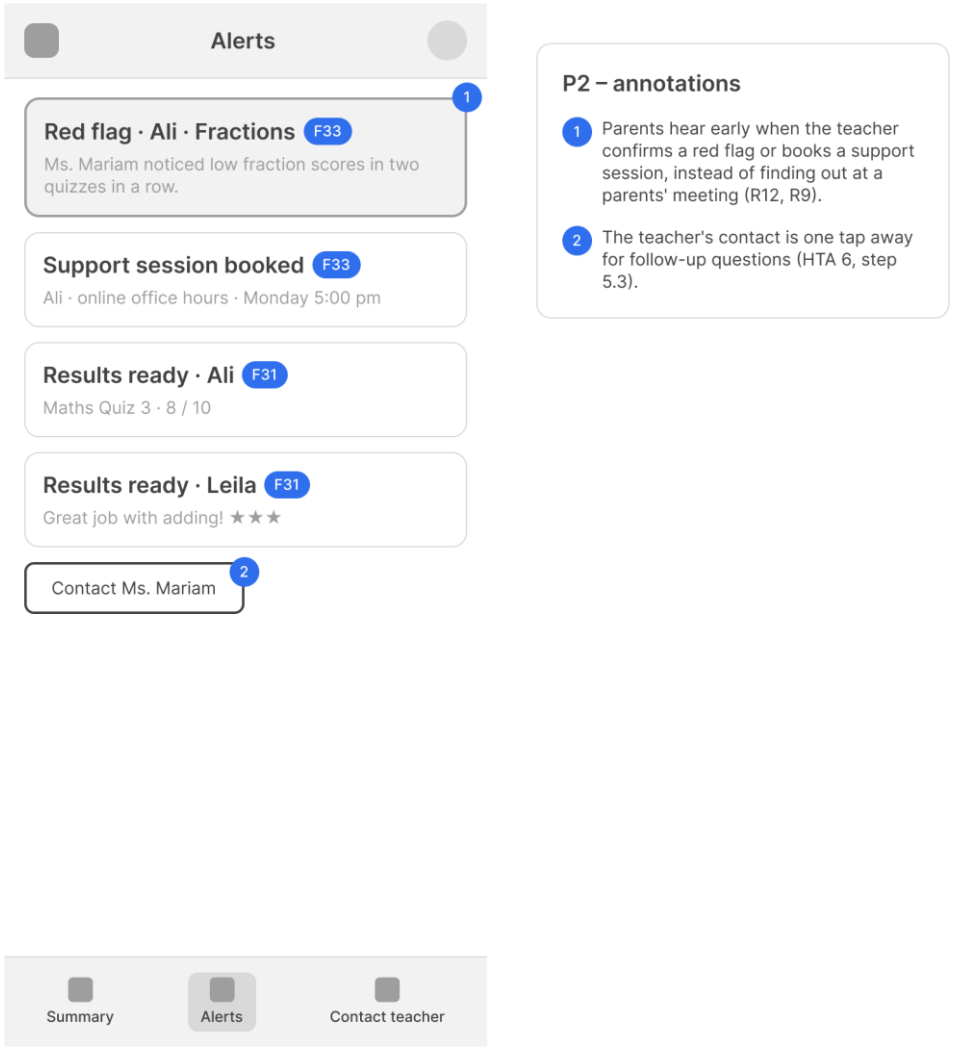


Figure 4.21 – P2 – Alerts – Parent

5. Teacher Interview

We interviewed a real mathematics teaching assistant using a written question-and-answer format. The questions focused on how the teacher grades exams, tracks students' weaknesses, and identifies students who need extra support, as required by the brief. The project requirements (R1–R16) below were collected from this interview.

5.1 Interview transcript

Item	Details
Interviewee	Nouran Ahmed
Role	High school mathematics teaching assistant
Format	Written question-and-answer transcript
Purpose	To understand how a real teacher currently grades exams, tracks student weaknesses, and identifies students who need extra support, to ground LearnLoop's personas and task analysis.

Q1. Can you walk me through how you grade a typical exam, from collecting the papers to returning results to students?

A: Physical exams/quizzes: After the papers are collected, they are corrected manually against a marking scheme and graded. The grades are entered into an Excel sheet, which is then converted into a parents' report stating that their child got X out of Y in this week's exam. The papers are returned to the students after 3–4 days, before the next week's quizzes.

Online exams/quizzes: After the students submit their papers on Google Classroom, the PDF is downloaded and corrected on a PDF editing platform against a marking scheme. The corrected version is then uploaded back to the student with the review. The grades are entered into an Excel sheet, which is then converted into a parents' report stating that their child got X out of Y in this week's exam. The corrected paper is returned to the student on Google Classroom after 2–4 days.

Q2. How do you currently tell the difference between a student who made a one-off mistake and one who has a repeated weakness in a specific topic, like fractions?

A: We check the homework, quizzes, and the questions they ask in general. If there is a consistent mistake, or they seem to grasp part of the concept but not all of it, we consider it a weakness.

Q3. Do you keep track of a student's mistakes over time, and if so, how? For example, notes, a spreadsheet, or just memory.

A: Each of us has our own method. Personally, I keep a notes document with the students' names and their problem area(s), which I update weekly after each week's quizzes and homework.

Q4. How do you decide when a student needs extra support, and how do you usually act on that decision?

A: It is decided in one of two ways. Either the students come to me or send me a message saying they do not understand or are struggling with a certain topic, or I reach out to them if their marks are consistently low for two weeks in a row. To help them, we schedule an online office-hours meeting so I can re-explain the concept and we can solve some questions together.

Q5. How do you currently share grades and feedback with students, and separately with parents?

A: Students: Students receive their mark on their corrected paper physically, or as a Google Classroom notification if they took the exam/quiz online.

Parents: Parents receive a written report with the following content:

Dear Parent,

This week's quiz mark for [Student Name] is X/Y.

Best regards,

Q6. If an AI tool read and graded a student's handwritten answers for you, what would you personally want to check before trusting its result?

A: I would check that everything is marked as correctly as if I had corrected it manually against the marking scheme.

Q7. Some students mix languages in their answers or have messy handwriting. How do you currently handle answers that are unclear or written in more than one language?

A: Messy handwriting: If I really cannot understand what is written, I consult a colleague. If they can understand it, that's fine. If not, I consider the question unanswered. Some students do clarify what they wrote afterwards, but they do not get the marks for that question back.

Mixed languages: As a math teacher, I do not face this often. When I do, the exam specifies that a particular question, or the whole exam/quiz, must be answered in one of the two languages.

Q8. What part of grading and tracking student progress is the most time-consuming or frustrating, and what would make the biggest difference for you?

A: Most frustrating: Flipping between the paper being graded and the marking scheme, entering the grades into the spreadsheet, and updating the progress spreadsheet.

Biggest difference: Something that automatically enters the grades into the sheet and returns them to the students as a notification, automatically updates the progress spreadsheet, and tells me what percentage is still unchecked or ungraded.

Q9. If you picture a dashboard for a system like this, what information would you want to see first? Do you have any preferences for its layout or colours?

A: Information: Statistics on how much of the updating and correction is done, the common problem areas for most of the students, and the students' names and contact information, as well as their parents'.

Layout and colours: I do not have a colour preference, but nothing too dark. I would like the layout to be clean, easy, and simple to navigate.

5.2 Key findings

Grading, tracking and reporting are manual and spread across paper, PDFs, Excel and personal notes. The teacher's biggest wish is automation plus a clear view of how much work is still unchecked.

1. **Grading is manual and fragmented.** Physical exams are marked by hand against a marking scheme. Online exams are downloaded from Google Classroom, marked in a PDF editor, and re-uploaded. Grades are typed into Excel and turned into a parent report. Results return 2–4 days after the exam. (Q1)
2. **Weaknesses are judged across several sources.** The teacher looks at homework, quizzes, and students' questions. A weakness is a consistent mistake, or partial understanding of a concept. (Q2)
3. **Tracking is personal and manual.** Each teacher has their own method. The interviewee keeps a notes document of each student's problem areas, updated weekly. (Q3)
4. **Extra support has two triggers.** Either the student reaches out, or the teacher reaches out after consistently low marks two weeks in a row. Support means an online office-hours session to re-explain and practise. (Q4)
5. **Communication is minimal.** Students get their mark on the paper or as a Google Classroom notification. Parents get only "This week's quiz mark is X/Y", with no feedback or weaknesses. (Q5)

6. **Trust in AI means full verification.** The teacher would check that everything is marked as they would have marked it against the marking scheme. (Q6)
7. **Unreadable handwriting is resolved through people.** The teacher asks a colleague. If no one can read it, the question counts as unanswered. Some students clarify what they wrote afterwards, but they do not get those marks back. (Q7)
8. **Mixed languages are rare and prevented by policy.** In math this seldom happens. When it does, the exam specifies one language per question or per exam. (Q7)
9. **The biggest pain points are switching and data entry.** Flipping between the paper and the marking scheme, entering grades into the spreadsheet, and updating the progress spreadsheet. (Q8)
10. **The teacher wants automation and visibility.** Grades entered automatically and sent to students as notifications, the progress tracker updated automatically, and the percentage still unchecked or ungraded shown. (Q8)
11. **Dashboard priorities and style.** First: correction progress, common class problem areas, and student and parent contact details. Style: light, not too dark, clean, simple to navigate. (Q9)

5.3 Derived requirements

Each requirement traces to an interview answer (Q-number). R-numbers are referenced by the features and wireframe annotations.

ID	Interview finding (source)	Derived requirement	LearnLoop feature	View
R1	Flipping between paper and marking scheme is frustrating (Q8)	Show the student's answer, the scan, and the marking scheme together	Side-by-side verification and grading screen	Teacher
R2	Teacher would check everything against the marking scheme before trusting AI (Q6)	Teacher can inspect and override every grade, including auto-accepted ones	Confirm or override with marking scheme visible; auto-accepted answers expandable at any time	Teacher
R3	Wants to see the percentage left unchecked or ungraded (Q8)	Show review progress at exam and class level	CGV time-saved and trust score; grading progress bar on dashboard	Teacher
R4	Grades typed manually into Excel (Q1, Q8)	Confirmed grades are recorded automatically	Auto-recorded gradebook with Excel export	Teacher
R5	Weekly manual notes on each student's problem areas (Q3)	Track each student's weaknesses over time automatically	Per-student weakness tracker	Teacher, Student
R6	Weakness = consistent mistake or partial understanding (Q2)	Detect recurring errors across exams, not only within one	Recurring-error detection and learning-gap view	Teacher
R7	Teacher reaches out after two weeks of low marks (Q4)	Flag sustained low performance for the teacher to review	Red flag rule (low score in a topic across two consecutive exams); confirm or dismiss	Teacher
R8	Students reach out when they don't understand (Q4)	Students can ask for help directly from their results	"I need help" button on feedback	Student
R9	Support = online office hours to re-explain (Q4)	Teacher can act on a confirmed flag immediately	Schedule a support session from a red flag	Teacher, Student
R10	Unreadable answers count as unanswered; clarifying later does not restore the marks (Q7)	Resolve unreadable answers before the grade is final, so students don't lose marks for handwriting alone	Ask the student to clarify a low-confidence answer, or consult a colleague, before grades are released	Teacher, Student
R11	Mixed languages rare; teacher sets one language (Q7)	Teacher sets the expected language; mismatches are surfaced	Per-exam or per-question language setting with mismatch flag	Teacher

ID	Interview finding (source)	Derived requirement	LearnLoop feature	View
R12	Parents receive only X/Y (Q5)	Parents see confirmed weaknesses and progress, not just a mark	Parent view with weakness alerts and progress summary	Parent
R13	Students get only a mark, 2–4 days later (Q1, Q5)	Students are notified as soon as results are confirmed, with feedback	Instant notification with grade, feedback, and weaknesses	Student
R14	Common class problem areas wanted first (Q9)	Show class-level weaknesses on the dashboard	Class weakness overview, next to the CGV class-wide OCR report	Teacher
R15	Student and parent contact details wanted on dashboard (Q9)	Quick access to student and parent contacts	Student roster with contact info	Teacher
R16	Light, clean, simple layout preferred (Q9)	Light theme, minimal navigation	Visual design constraint for wireframes and prototype	All

5.4 Implications for LearnLoop

What the interview supports:

- CGV trust score and progress. The teacher asked unprompted for the percentage left unchecked (R3), which matches CGV's "checked by hand vs. auto-accepted" indicator.
- Automated weakness tracking. The weekly notes document and progress spreadsheet (R4, R5) are exactly the manual work LearnLoop replaces.
- Human-in-the-loop design. The teacher wants to verify results against the marking scheme (R2), confirming that the teacher must stay in control.
- Parent view. Parents currently receive only X/Y (R12), so a parent view adds clear value.

Adjustments to the Milestone 1 proposal:

- Auto-accepted answers stay visible on demand. Because the teacher wants to check everything (R2), auto-accepted answers are collapsed rather than hidden, expandable in one click, and the teacher can adjust the confidence threshold.
- English and German exam support (F11) stays a core requirement. Only mixed-language answers (both languages within one answer) become a lighter feature. The interview showed they are rare in maths and handled by policy (R11), so instead of a separate tool, the teacher sets the expected language and CGV sends any answer in a different language to the "needs a decision" tier for review.

New features that came from the interview:

- Student clarification request (R10): an unreadable answer in CGV's "needs a decision" tier can be sent to the student to clarify before grades are released.
- "I need help" button (R8) and support-session scheduling (R9): these digitise both support triggers.
- Side-by-side marking scheme (R1): addresses the most frequently mentioned frustration.

Limits of this interview:

- The interview did not cover how young children react to grades and feedback; the student personas state these as assumptions.
- The teacher did not comment on confidence scores specifically, so the interview supports CGV's progress indicator but does not yet validate confidence tiers.

AI Use Statement and Chat History

As required by the course, we disclose our use of AI tools (Claude and ChatGPT) in this milestone. We used them to track our progress against the brief, suggest structure and draft wording for the personas, HTAs and feature list, review our drafts against the course requirements, and generate the HTA diagrams, the wireframe kit and the wireframe screens in Figma from our own specifications.

All AI output was reviewed, edited and approved by the team, and design decisions (such as the parent view and the effort-points features) were made by the team. Interview findings come only from the real teacher interview.

Full chat history

- Claude conversation 1
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- Claude conversation 2
- ChatGPT conversation